

A counterexample to Ryser’s covering conjecture

OpenAI

September 23, 2026

Abstract

For every sufficiently large prime $s \equiv 2 \pmod{3}$ and every sufficiently large odd integer n , with the threshold depending on s , we construct an intersecting $(s^n + 1)$ -partite $(s^n + 1)$ -uniform hypergraph with covering number $s^n + 1$. This disproves Ryser’s covering conjecture in its intersecting case.

1 Introduction

A vertex cover of a finite hypergraph H is a set of vertices meeting every edge. Its covering number $\tau(H)$ is the minimum size of such a set, and its matching number $\nu(H)$ is the maximum number of pairwise disjoint edges. For an integer $r \geq 2$, an r -partite r -uniform hypergraph has disjoint vertex parts $V_1, \dots, V_r$, with every edge containing exactly one vertex from each part. Ryser’s covering conjecture asserts that every such hypergraph satisfies

$$\tau(H) \leq (r - 1)\nu(H). \quad (1)$$

A nonempty hypergraph is intersecting when every two distinct edges meet; equivalently, its matching number is one. In this case the conjecture asks for a cover of size at most $r - 1$. The vertices of a cover may lie in several different parts, which is the central obstacle in constructing a counterexample.

For $r = 2$, inequality (1) is König’s matching theorem. An equivalent formulation appears in Henderson’s 1971 thesis [22]; Best and Wanless discuss the history of its attribution to Ryser [6]. Aharoni proved the full tripartite case [3]. In the intersecting case, the results through rank four are due to Gyárfás, and the rank-five result to Tuza, as recorded by Király and Tóthmérész [24]; see also Tuza’s early study of the covering conjecture [30]. Haxell and Scott proved $\tau(H) \leq (r - \epsilon)\nu(H)$ for some $\epsilon > 0$ when $r = 4$ or 5 , without assuming intersection [20]. White proved the particular rank-four case $\nu(H) = 2$, obtaining $\tau(H) \leq 6$ [31].¹ Under the additional assumption that any two distinct edges meet in exactly one vertex, Francetić, Herke, McKay, and Wanless established the intersecting case through rank nine [15]. Bishnoi, Das, Morris, and Szabó obtained bounds when every pair of edges has a larger prescribed intersection [7].

Finite planes supply the classical equality examples. Delete a point and all lines through it from a projective plane of order q . Partition the remaining points according to the deleted line through them, and take the remaining lines as hyperedges. The result is intersecting, has $q + 1$ parts and covering number q . Thus equality in the proposed bound occurs whenever $r - 1$ is a prime power. Work of Mansour, Song, and Yuster [25] and Aharoni, Barát, and Wanless [4] studies the structure and sparsity of equality examples. Abu-Khazneh, Barát, Pokrovskiy, and Szabó exploit the rigidity of minimum covers in a truncated plane and add exceptional edges to obtain further

¹White reports assistance from GPT-5.6 Sol and Claude in the Methods section of his preprint.

equality examples [2]. In that construction, the number of parts and the covering number both increase by one. Haxell and Scott combine affine planes with intersecting configurations to obtain covering number at least $r - 4$ for every sufficiently large r , improved to $r - 3$ when r is even [21].

Our construction also starts from a finite plane and inserts exceptional edges, but keeps one part for each ambient direction. It changes the line labels within these parts to raise the covering number beyond q . The following theorem gives the precise field orders for which this can be done.

Theorem 1.1. *There exists s_0 such that, for every prime $s \geq s_0$ with $s \equiv 2 \pmod{3}$, there is an integer $n_0(s) \geq 3$ with the following property. For every odd integer $n \geq n_0(s)$, put $q = s^n$. There is a finite intersecting $(q + 1)$ -partite $(q + 1)$ -uniform hypergraph H satisfying*

$$\tau(H) = q + 1.$$

In particular, the intersecting case of (1) is false.

Every edge of a nonempty intersecting hypergraph is itself a cover, so the assertion $\tau(H) \leq q + 1$ is automatic. The proof must construct an intersecting hypergraph and exclude every cover of size at most q . We first fix any sufficiently large eligible prime s , construct a configuration in $\mathbb{F}_s^2$, and then take n sufficiently large among the odd integers. All subsequent thresholds may depend on that fixed configuration. The result is asymptotic; the proof does not give a numerical threshold.

A companion paper [27] proves a different theorem: for every sufficiently large prime q , it constructs such a hypergraph with exactly $q + 1$ vertices in every part and with every vertex nonisolated. Its construction modifies line partitions directly in the prime plane. Here the field order is a proper odd power, and the proof uses a small-field configuration with a nonlinear change of coordinates. The two constructions give distinct ranges of ranks. The only results imported from the companion are its three elementary estimates in Appendix A: the affine second-moment identity (Lemma A.1), bounded-drift concentration (Lemma A.2), and the exposure bound for independent trials (Lemma A.3).

Lovász proposed the stronger assertion that deleting $r - 1$ vertices can always reduce the matching number. Clow, Haxell, and Mohar disproved that assertion at $r = 3$ [11]; Abiad, Garbe, Povill, and Spiegel subsequently gave an infinite family at $r = 3$ and examples at $r = 4$ [1]. These examples are consistent with (1). They exclude a prescribed matching-reduction procedure, whereas Theorem 1.1 violates the covering inequality itself.

Monochromatic tree covers

An r -edge-coloring here assigns exactly one of r colors to each edge. A monochromatic tree cover is a family of monochromatic trees whose vertex sets cover the graph; the trees may overlap, and singleton trees are allowed. Gyárfás's conjecture asks whether $r - 1$ such trees suffice for every r -edge-colored complete graph; see [26, Conjecture 1]. This is the covering formulation recorded after Theorem 2 in [13, Section 3], not the distinct vertex-disjoint tree-partition conjecture numbered Conjecture 2 there.

Corollary 1.2. *Let s_0 and $n_0(s)$ be as in Theorem 1.1. For every prime $s \geq s_0$ with $s \equiv 2 \pmod{3}$ and every odd integer $n \geq n_0(s)$, put $r = s^n + 1$. There is a finite simple complete graph with an r -edge-coloring for which the minimum number of monochromatic trees needed to cover its vertices is exactly r . The same minimum holds with connected monochromatic subgraphs or maximal monochromatic components in place of trees.*

Proof. Take the hypergraph H from Theorem 1.1, with parts $V_1, \dots, V_r$. For $e \in E(H)$, let $x_i(e)$ be the unique vertex in $e \cap V_i$. Form the complete graph K with vertex set $E(H)$. For each pair of distinct hyperedges e, f , choose one index i with $x_i(e) = x_i(f)$ and color the edge ef by i . Such an index exists because H is intersecting. Only one color is chosen even when the two hyperedges meet in several parts.

Along a color- i path, the value of x_i is constant. Thus every connected color- i subgraph determines a vertex of V_i belonging to every hyperedge indexed by that subgraph. For a singleton, choose any color and the corresponding vertex of its hyperedge. If t connected monochromatic subgraphs cover the vertices of K , then at most t hypergraph vertices so chosen meet every edge of H . Hence $t \geq \tau(H) = r$.

Conversely, fix a vertex of K . The r monochromatic stars centered there, one for each color and allowing singleton stars, cover all vertices of K . They give a tree cover of size at most r . The lower bound applies to trees and to maximal monochromatic components as well, and the stars can be enlarged to maximal components. This proves all three stated minima. $\square$

Thus Gyárfás's cover conjecture fails at these extension-field ranks. The examples admit no cover by $r - 1$ connected monochromatic subgraphs even without a diameter restriction. They therefore also refute Milićević's bounded-diameter strengthening [26, Conjecture 6], whose diameter bound may depend on r , and the $\alpha = 1$ case of the conjecture of DeBiasio, Kamel, McCourt, and Sheats [12, Conjecture 2.8], whose diameter bound depends only on the independence number α . Indeed, $\alpha(K) = 1$, so the latter conjecture also demands a cover by at most $r - 1$ such subgraphs.

Proof strategy

In the affine-plane model, the parts are indexed by the $q + 1$ directions in $\mathbb{F}_q^2$, and the vertices of a part are the q lines of that direction. A point x defines an edge consisting of its incident line in each direction. The edges of two points intersect at their join, but a full parallel class always gives a cover of size q . We retain many of these point edges and add exceptional edges whose labels are changed in some directions. The construction has three tasks: preserve intersection, restrict small covers of the retained point edges, and make the exceptional edges too costly for those covers.

Restricting the ordinary cover. Delete a bounded number of translated shapes in each of $O(q)$ independent trials. The shapes are lines or curves that meet every line in boundedly many points. Thus a shape has bounded intersection with a line unless it is that line itself. Section 3 proves that, under explicit uniform survival assumptions, every cover of almost all surviving points by at most q lines contains almost a pencil. Here a pencil is a family of lines through one projective point, including a parallel class when that point is at infinity. Surviving points on each nondeleted line then force every nondeleted member of this pencil into the cover. To obtain the pencil statement over the required extension fields, Section 2 first proves an incidence saving by the sum-product method of Bourgain, Katz, and Tao [9]. Its constants are uniform over fields whose proper subfields all have order at most $q^{1/3}$.

One configuration in two coordinates. In a fixed prime plane $\mathbb{F}_s^2$, remove one line in each finite nonzero direction and call the remaining set Y . The change of coordinates $f(x, y) = (x^3, y)$ is bijective when $s \equiv 2 \pmod{3}$. Original line labels determine which ambient point edges must be deleted to intersect the exceptional edges. The labels for altered directions are instead read from the lines through $f(w)$, for $w \in Y$. The map f preserves the horizontal and vertical parallel classes; this ensures intersection within the exceptional configuration. Its nonlinear action on the

other lines makes their images curves with at most three intersections with any line. Applying the deletion lemma in these new coordinates yields the crucial obstruction: covering $f(Y)$ by a lines and b individual points with $a + b/2 \leq s$ requires exactly one full parallel class and no point resources. Section 5 proves this statement together with the ordinary-coordinate occupancy needed later.

Placing the configurations and spending the cover budget. Embed linearly transformed copies of Y in $\mathbb{F}_q^2$ so that every ambient direction is altered in one or two copies. Arbitrary translates could create disjoint exceptional edges. We therefore sample a fixed number of independent candidate translates for each copy. Section 4 proves that one can choose compatible representatives from these pools, using an independent-transversal argument in the tradition of Haxell [18, 19]. Section 6 proves the deletion estimates for the entire independent pool before making this selection. Keeping only the selected deletions restores ordinary points, so the needed estimates transfer by deterministic inclusion.

Section 7 gives the labels of every edge, proves intersection, and excludes a cover of size at most q . After the ordinary edges force a pencil, its center determines the remaining argument. At infinity, the uncovered exceptional configurations require more vertices than the missing parallel lines release. Inside an embedded configuration, the remaining budget would cover $f(Y)$ by at most s lines including two nonparallel axis lines. Outside every configuration, each deleted pencil line supplies five uncovered exceptional edges, while every additional vertex covers at most five. In all three cases the available budget is too small.

Conventions. A direction is a one-dimensional linear subspace of $\mathbb{F}_q^2$; as a subspace it includes zero. The projective completion adds one point at infinity per direction. A pencil consists of the affine lines through a projective point: it has $q + 1$ members for a finite center and q members for a center at infinity. Thus concurrence always includes parallel lines. For a positive integer j , write $(a)_j = a(a - 1) \cdots (a - j + 1)$. All asymptotic constants may depend on the parameters explicitly held fixed. An event holds with high probability if its probability tends to one along the specified sequence of fields. The small field is fixed before the ambient field tends to infinity.

2 A uniform incidence bound

For a set P of points and a set $\mathcal{L}$ of lines in $\mathbb{P}^2(\mathbb{F}_q)$, write

$$I(P, \mathcal{L}) = |\{(p, \ell) \in P \times \mathcal{L} : p \in \ell\}|. \quad (2)$$

Our aim is a power saving with constants independent of the field and its characteristic. In Section 3, this estimate will control intersections of a proposed line cover when no projective point lies on many of its lines. This is the incidence input to the conclusion that a small cover must contain almost a full pencil.

Theorem 2.1 (Uniform incidence saving). *There exist absolute constants $\kappa \in (0, 1/10)$ and Q with the following property. Suppose that $q \geq Q$ is a prime power and every proper subfield of $\mathbb{F}_q$ has order at most $q^{1/3}$. If P and $\mathcal{L}$ are sets of projective points and projective lines over $\mathbb{F}_q$ satisfying*

$$|P|, |\mathcal{L}| \leq q^{1+\kappa}, \quad (3)$$

then

$$I(P, \mathcal{L}) \leq q^{3/2-\kappa}. \quad (4)$$

The proof follows the incidence and sum-product strategy of Bourgain, Katz, and Tao [9, Sections 3–4 and 6]. Its geometric step turns an incidence configuration with too many incidences into a dense set of pairs whose differences and quotients range over small sets. A graph argument then extracts one subset with both a small difference set and a small quotient set. Finally, scalar coverings show that such a subset cannot have size near the square root of the field order under the proper-subfield hypothesis.

We develop the two algebraic ingredients first, then return to the incidence configuration. The proofs include the simultaneous extraction of additive and multiplicative control and the bounds on scalar expressions that make the argument uniform over fields of nonprime order.

2.1 Finite-set estimates

The first task is to pass from control of differences and quotients along the edges of a dense bipartite graph to control on all pairs in one subset. We begin with the sumset estimates needed to use the resulting additive control.

For subsets of an abelian group, $X + Y$ and $X - Y$ denote the usual sumset and difference set. For a nonnegative integer j , the notation jY in this subsection means a sum of j copies of Y , with $0Y = \{0\}$. Later, a field element multiplying a set will always mean field scaling; we will specify explicitly when an integer denotes an iterated sumset.

The first two estimates below are Ruzsa’s triangle inequality and covering lemma; the maximal-translates proof of the latter appears in [29, p. 324]. The multiple-sum bound is a Plünnecke–Ruzsa estimate, for which we use Petridis’s minimizing-subset proof [28, Proposition 2.1 and Section 3].

Lemma 2.2 (Triangle, covering, and multiple-sum estimates). *Let the sets below be finite subsets of an abelian group.*

1. *If Z is nonempty, then*

$$|X - Y| |Z| \leq |X - Z| |Z - Y|. \quad (5)$$

2. *If Y is nonempty and $|X - Y| \leq M|Y|$, then X is contained in the union of at most M translates of $Y - Y$.*

3. *If Y is nonempty and $|Y - Y| \leq M|Y|$, then, for all nonnegative integers u, v ,*

$$|uY - vY| \leq M^{u+v} |Y|. \quad (6)$$

Proof. For the triangle inequality, choose one representation $d = x_d - y_d$ for each $d \in X - Y$. The map

$$(d, z) \mapsto (x_d - z, z - y_d)$$

from $(X - Y) \times Z$ into $(X - Z) \times (Z - Y)$ is injective: summing the two coordinates recovers d , after which the chosen representation recovers z . This proves (5).

For the covering statement, take a maximal set $T \subseteq X$ for which the translates $t - Y$, $t \in T$, are pairwise disjoint. Their union is contained in $X - Y$, so $|T| \leq M$. Maximality implies that for every $x \in X$ some $x - Y$ intersects $t - Y$ with $t \in T$, whence $x \in t + (Y - Y)$.

For the multiple-sum bound, the minimizing-subset argument of Petridis [28] lets us iterate a one-step growth estimate. Put $Q_0 = -Y$ and choose a nonempty $W \subseteq Y$ minimizing

$$\mu = \frac{|W + Q_0|}{|W|}.$$

Then $\mu \leq M$, and every $W' \subseteq W$, including the empty set, satisfies $|W' + Q_0| \geq \mu|W'|$. We claim that for every finite set E_0 ,

$$|W + Q_0 + E_0| \leq \mu|W + E_0|. \quad (7)$$

The assertion is immediate if E_0 is empty. Otherwise order its elements as $e_1, \dots, e_k$. When the translate $W + e_i$ is added to the preceding translates, let

$$W'_i = \{w \in W : w + e_i \in \bigcup_{j < i} (W + e_j)\}.$$

It contributes exactly $|W| - |W'_i|$ new elements. Moreover, $W'_i + Q_0 + e_i$ is contained in the union of the preceding enlarged translates $W + Q_0 + e_j$. Thus the contribution of $W + Q_0 + e_i$ is at most

$$|W + Q_0| - |W'_i + Q_0| \leq \mu(|W| - |W'_i|).$$

Summing proves (7). Iterating this estimate gives $|W + jQ_0| \leq \mu^j|W|$ for every nonnegative integer j . We now remove the auxiliary set W by applying (5) to uQ_0, vQ_0 with intermediate set $-W$:

$$|uQ_0 - vQ_0| |W| \leq |W + uQ_0| |W + vQ_0| \leq \mu^{u+v} |W|^2.$$

Negation preserves cardinality, so $|uQ_0 - vQ_0| = |uY - vY|$. Since $|W| \leq |Y|$ and $\mu \leq M$, this is (6). $\square$

We next carry out the extraction from a dense graph. The argument uses the common-neighborhood method in Gowers's quantitative refinement [17, Section 4, Lemma 11 and Proposition 12] of the Balog–Szemerédi theorem [5]; see also [14, Section 5, Lemmas 5.1–5.2]. The point is to use the same walks for both additive and multiplicative edge labels. We can therefore obtain both bounds on one subset, which is what the subsequent subfield argument requires. In the geometric application, the edge labels will come from the four pencils through four chosen points.

Lemma 2.3 (Simultaneous difference and quotient extraction). *Let F be a field, let $A, B \subseteq F^*$ be finite, and let $G \subseteq A \times B$. Define*

$$D = \{a - b : (a, b) \in G\}, \quad J = \{a/b : (a, b) \in G\}.$$

Suppose that, for real numbers $n > 0$ and $V \geq 2$,

$$|A|, |B|, |D|, |J| \leq Vn, \quad |G| \geq n^2/V. \quad (8)$$

Then there is $S \subseteq A$ such that

$$\frac{3n}{8V^2} \leq |S| \leq Vn \quad (9)$$

and

$$|S - S|, |S/S| \leq 2^{20} V^{22} n. \quad (10)$$

Here $S/S = \{a/a' : a, a' \in S\}$.

Proof. We first find a large set of vertices such that every ordered pair is joined by many walks of length four. Counting the possible edge labels of these walks will then bound both differences and quotients of the endpoints.

Regard G as a bipartite graph, and put $\tau = (512V^8)^{-1}$. Call an ordered pair $(a, a_2) \in A^2$ bad if it has fewer than τn common neighbors in B . To find a neighborhood with few bad pairs, first note that vertices of B with fewer than $n/(2V^2)$ neighbors account for at most $|B|n/(2V^2) \leq n^2/(2V)$

edges. Since every vertex of B has at most Vn neighbors, the set of vertices with at least $n/(2V^2)$ neighbors has size at least $n/(2V^2)$.

Summed over all $b \in B$, the number of bad ordered pairs in the neighborhood of b is at most

$$|A|^2 \tau n \leq V^2 \tau n^3.$$

Averaging over the large neighborhoods just identified, one obtains a neighborhood A_0 of size at least $n/(2V^2)$ in which the number of bad ordered pairs is at most

$$2V^4 \tau n^2 = \frac{n^2}{256V^4} \leq \frac{|A_0|^2}{64} \leq \frac{|A_0|^2}{16}.$$

Let S consist of those $a \in A_0$ having at most $|A_0|/4$ bad partners in A_0 . At most $|A_0|/4$ elements are discarded, so $|S| \geq 3|A_0|/4$, proving (9).

The choice of S ensures that, for every $a, a' \in S$, at least $|A_0|/2$ elements $a_2 \in A_0$ form nonbad pairs with both endpoints. Each such a_2 gives at least $(\tau n)^2$ ordered walks

$$a, b_1, a_2, b_2, a'$$

whose four edges belong to G . Repeated vertices are allowed. Consequently each ordered endpoint pair in S^2 has at least

$$\frac{|A_0|}{2} (\tau n)^2 \geq \frac{n^3}{2^{20} V^{18}} \quad (11)$$

such walks. This lower bound now converts the limited number of edge labels into a bound on endpoint differences.

For one of these walks, write its edge differences, always taking the A coordinate minus the B coordinate, as

$$d_1 = a - b_1, \quad d_2 = a_2 - b_1, \quad d_3 = a_2 - b_2, \quad d_4 = a' - b_2.$$

They satisfy $a - a' = d_1 - d_2 + d_3 - d_4$. For a fixed endpoint pair, the four differences determine the walk uniquely: starting from a , one recovers successively b_1, a_2, b_2, a' . Choose one ordered endpoint pair for each element of $S - S$. The resulting collections of difference quadruples are disjoint for different differences, because their alternating sums differ. Counting in D^4 and using (11) gives

$$|S - S| \leq \frac{|D|^4}{n^3/(2^{20} V^{18})} \leq 2^{20} V^{22} n.$$

The same walks give the quotient bound on the same set S . Put

$$j_1 = a/b_1, \quad j_2 = a_2/b_1, \quad j_3 = a_2/b_2, \quad j_4 = a'/b_2.$$

All are nonzero, and $a/a' = j_1 j_3 / (j_2 j_4)$. From a fixed initial a and these four quotients one recovers $b_1 = a/j_1$, $a_2 = j_2 b_1$, $b_2 = a_2/j_3$, and $a' = j_4 b_2$. Choosing one endpoint pair for each element of S/S and counting in J^4 therefore proves the other bound in (10). $\square$

2.2 Bounded scalar expressions and subfields

We now turn to the algebraic obstruction: a set near the square-root field scale cannot have both the small difference set and the small quotient set supplied by the extraction lemma. The first step is to bound sums involving scalars formed using a bounded number of field operations. We use the scalar-covering method of Bourgain, Katz, and Tao [9, Lemma 3.2 and Proposition 3.3], keeping the dependence on the number of operations explicit.

An expression in a set $E \subseteq F$ is built from members of $E \cup \{0, 1\}$ using binary addition, subtraction, and multiplication, and unary negation and inversion. Each of these operations counts as one operation. Inversion is allowed only when its argument is nonzero.

Lemma 2.4 (Uniform scalar coverings). *Let B be a nonempty finite subset of a field F , let $E \subseteq B$, and let $K \geq 2$. Suppose that*

$$|B - B| \leq K|B|, \quad |B - eB| \leq 2K^3|B| \quad (e \in E). \quad (12)$$

Define

$$C_0 = 4, \quad C_{m+1} = 3C_m + 5, \quad \text{so that } C_m = \frac{13 \cdot 3^m - 5}{2}. \quad (13)$$

If $\lambda \in F$ has an expression in E using at most m operations, then λB is contained in at most K^{C_m} translates of $B - B$. In particular,

$$|E + \lambda E| \leq |B + \lambda B| \leq K^{C_m+3}|B|. \quad (14)$$

These bounds are uniform over all such expressions and all choices of their entries from E .

Proof. Put $H = B - B$, so $H = -H$ and $0 \in H$. In the remainder of this proof, integer multiples of B or H inside sumsets denote iterated sums. We prove the covering assertion by induction on expression length. To combine coverings at an induction step, we will need a fixed covering of $2H$ by translates of H . By Lemma 2.2,

$$|2H - B| = |2B - 3B| \leq K^5|B|.$$

The covering statement of that lemma consequently supplies a set $T_* \subseteq F$ with $|T_*| \leq K^5$ such that

$$2H \subseteq T_* + H. \quad (15)$$

Every $e \in E$ has a covering $eB \subseteq T_e + H$ with $|T_e| \leq 2K^3 \leq K^4$, by (12) and the covering lemma. The scalars 0 and 1 also have coverings by one translate of H : use $\{0\} \subseteq H$ and $B \subseteq b_0 + H$ for any $b_0 \in B$. This proves the case $m = 0$.

We now check how the number of translates changes under each allowed operation. Suppose that

$$\lambda B \subseteq T + H, \quad \mu B \subseteq U + H, \quad |T| \leq K^a, \quad |U| \leq K^b.$$

Negation replaces T by $-T$. For addition and subtraction,

$$(\lambda \pm \mu)B \subseteq \lambda B \pm \mu B \subseteq T \pm U + 2H \subseteq T \pm U + T_* + H,$$

so the number of translates is at most K^{a+b+5} . For a product,

$$\lambda\mu B \subseteq \lambda U + \lambda H = \lambda U + (\lambda B - \lambda B) \subseteq \lambda U + T - T + T_* + H.$$

This uses at most K^{b+2a+5} translates and is valid even if $\lambda = 0$. Finally, if $\lambda \neq 0$, then

$$|\lambda B - B| \leq |T| |H - B| = |T| |B - 2B| \leq K^{a+3}|B|.$$

Scaling by λ^{-1} and then negating gives $|\lambda^{-1}B - B| \leq K^{a+3}|B|$. The covering lemma now covers $\lambda^{-1}B$ by at most K^{a+3} translates of H .

These bounds give the claimed uniform induction: in an expression with at most $m+1$ operations, each immediate subexpression has at most m operations, so every resulting covering uses at most K^{3C_m+5} translates. This proves the covering assertion with (13).

To deduce the sumset bound from the covering, let $\lambda B \subseteq T + H$ be the covering just obtained. Then

$$|B + \lambda B| \leq |T| |B + H| = |T| |2B - B| \leq K^{C_m+3}|B|,$$

again by Lemma 2.2. Since $E \subseteq B$, this also bounds $|E + \lambda E|$. $\square$

When working along a sequence $q \rightarrow \infty$, a positive quantity $K = q^{o(1)}$ means that $\log K / \log q \rightarrow 0$. Likewise, $|S| = q^{1/2+o(1)}$ means that $\log |S| / \log q \rightarrow 1/2$. In particular, every fixed power of a subpower factor is still a subpower factor. This is why a uniform bound for expressions of fixed length suffices for the next lemma.

Lemma 2.5 (Exclusion of an intermediate approximate subfield). *There is no sequence of prime powers $q \rightarrow \infty$, each satisfying the proper-subfield hypothesis of Theorem 2.1, together with sets $S \subseteq \mathbb{F}_q^*$ and numbers $K \geq 2$ such that*

$$|S| = q^{1/2+o(1)}, \quad K = q^{o(1)}, \quad |S - S|, |S/S| \leq K|S|. \quad (16)$$

Proof. Suppose that such a sequence exists. We first use multiplicative overlaps to obtain sets satisfying the scalar-covering hypotheses. Ratios of differences in a large subset will then form a subfield. The proper-subfield hypothesis will force that subfield to be the whole field, where averaging will contradict the scalar-covering bound.

Extracting scalar-covering hypotheses. Put $s = |S|$. The s sets S/a , $a \in S$, each have size s , and their union is contained in S/S . Cauchy–Schwarz gives

$$\sum_{a,b \in S} |S/a \cap S/b| \geq \frac{s^4}{|S/S|} \geq \frac{s^3}{K}. \quad (17)$$

Thus some $b \in S$ satisfies $\sum_{a \in S} |S/a \cap S/b| \geq s^2/K$. Since every summand is at most s , at least $s/(2K)$ values $a \in S$ have $|S/a \cap S/b| \geq s/(2K)$: otherwise the sum of the large summands would be less than $s^2/(2K)$ and that of all the other summands at most $s^2/(2K)$.

Let E be the set of $e = a/b$ for these values of a , and put $B = S/b$. Then

$$E \subseteq B, \quad |B| = s, \quad |E| \geq \frac{s}{2K}, \quad |S \cap eS| \geq \frac{s}{2K} \quad (e \in E). \quad (18)$$

The last inequality follows by multiplying $S/a \cap S/b$ by a . For $Z = S \cap eS$, the triangle inequality and the containments $Z \subseteq S, eS$ imply

$$|S - eS| \leq \frac{|S - Z| |Z - eS|}{|Z|} \leq \frac{(Ks)^2}{s/(2K)} = 2K^3s.$$

After scaling by $1/b$, we have

$$|B - B| \leq Ks, \quad |B - eB| \leq 2K^3s \quad (e \in E).$$

Thus Lemma 2.4 applies to B and E . We have obtained uniform control for every scalar expressible using a bounded number of operations on elements of E .

Ratios of differences form a subfield. Write $t = |E|$. By (18) and (16), $t = q^{1/2+o(1)}$. In particular, $t \geq 2$ for all sufficiently large fields in the sequence. Define

$$R_0 = \left\{ \frac{x - x'}{y - y'} : x, x', y, y' \in E, y \neq y' \right\}. \quad (19)$$

To prove closure under addition and multiplication, we apply the scalar-covering bound to the result of either operation. Every member of R_0 has an expression in E with at most four operations.

Consequently, for $r, r' \in R_0$, each of $r + r'$ and rr' has an expression with at most nine operations. For either candidate λ , Lemma 2.4 gives

$$|E + \lambda E| \leq K^{C_9+3}s < t^2$$

for all sufficiently large q , uniformly over r, r' . Indeed, the ratio of this upper bound to t^2 is at most $4K^{C_9+5}/s$, which tends to zero. Therefore the map $(x, y) \mapsto x + \lambda y$ from E^2 is not injective. A collision between distinct pairs satisfies $y \neq y'$, since otherwise also $x = x'$, and gives

$$\lambda = \frac{x' - x}{y - y'} \in R_0.$$

It follows that R_0 is closed under addition and multiplication. It contains 0 and 1 and, directly from its definition, is closed under negation and reciprocals of nonzero elements. Thus R_0 is a subfield of $\mathbb{F}_q$.

The size of E now prevents this subfield from being proper. Choose distinct $e_0, e_1 \in E$. The map

$$e \mapsto \frac{e - e_0}{e_1 - e_0}$$

injects E into R_0 , so $|R_0| \geq t > q^{1/3}$ for large q . The proper-subfield hypothesis forces $R_0 = \mathbb{F}_q$.

A scalar with a large sumset. Every field element consequently has a four-operation representation using elements of E . The scalar-covering estimate therefore bounds $|E + \lambda E|$ for every scalar. To reach a contradiction, we average over the field and find a scalar for which this sumset is too large.

For $\lambda \in \mathbb{F}_q$, let N_λ be the number of ordered quadruples $(x, y, x', y') \in E^4$ satisfying $x + \lambda y = x' + \lambda y'$. There are t^2 identical pairs, which give a collision for every λ . If $y \neq y'$, exactly one value of λ gives a collision; if $y = y'$ but $x \neq x'$, none does. Hence

$$\frac{1}{q} \sum_{\lambda \in \mathbb{F}_q} N_\lambda = t^2 + \frac{t^4 - t^3}{q} \leq t^2 + \frac{t^4}{q}. \quad (20)$$

For some λ , Cauchy–Schwarz applied to the fibers of $(x, y) \mapsto x + \lambda y$ therefore yields

$$|E + \lambda E| \geq \frac{t^4}{t^2 + t^4/q} \geq \frac{1}{2} \min(t^2, q) = q^{1-o(1)}. \quad (21)$$

But $R_0 = \mathbb{F}_q$, so this particular λ itself has a four-operation representation in (19). Lemma 2.4 bounds its sumset by $K^{C_4+3}s = q^{1/2+o(1)}$, contradicting (21). The expression length here remains bounded; no iteration of field generation is being used. $\square$

2.3 Proof of the incidence theorem

The algebraic ingredients are now in place. It remains to obtain the hypotheses of Lemma 2.3 from a configuration with too many incidences. We use the projective-coordinate reduction in the incidence strategy of Bourgain, Katz, and Tao [9, Section 6]: after controlling degrees, we choose four points whose pencils encode the two coordinates, their difference, and their quotient.

Proof of Theorem 2.1. We argue by contradiction, allowing all quantitative losses to be subpower factors along a sequence of fields. Suppose the theorem is false. For a sequence of positive exponents $\kappa_j \rightarrow 0$ in $(0, 1/10)$, choose admissible counterexamples with $q_j \rightarrow \infty$. Such choices are possible because the failure of the theorem says that for each fixed exponent there are arbitrarily large counterexamples. Along this sequence, put $U = \max(2, q_j^{\kappa_j})$ and suppress the index j . We have

$$|P|, |\mathcal{L}| \leq Uq, \quad I(P, \mathcal{L}) \geq \frac{q^{3/2}}{U}, \quad U \geq 2, \quad U = q^{o(1)}. \quad (22)$$

All bounds below use absolute constants. Every assertion made for sufficiently large q is along this sequence. Our goal is to construct from these counterexamples sets of the size and small-expansion form excluded by Lemma 2.5.

Pruning the incidence graph. We first retain many incidences while putting both an upper and a lower bound on every remaining degree. Consider the bipartite graph on P and $\mathcal{L}$ whose edges are incidences. If $d(v)$ denotes a degree on either side, then counting ordered pairs on the opposite side and using the unique join of two distinct projective points, or the unique intersection of two distinct projective lines, gives

$$\sum_v d(v)^2 \leq 2U^2q^2 \quad \text{on each side.} \quad (23)$$

Here $\sum_v d(v)(d(v) - 1) \leq U^2q^2$, and $\sum_v d(v) = I(P, \mathcal{L}) \leq U^2q^2$.

Set

$$H = 16U^3q^{1/2}, \quad d = \frac{q^{1/2}}{8U^2}. \quad (24)$$

Delete every vertex whose initial degree exceeds H . By (23), the total number of incidences lost, counting both sides, is at most

$$\frac{4U^2q^2}{H} = \frac{q^{3/2}}{4U}.$$

Next, iteratively delete vertices whose current degree is less than d . Each of the at most $2Uq$ original vertices is deleted at most once, so this stage loses at most $2Uqd = q^{3/2}/(4U)$ incidences. Denote the surviving sets again by $P, \mathcal{L}$. They satisfy

$$I(P, \mathcal{L}) \geq \frac{q^{3/2}}{2U}, \quad \frac{q}{32U^4} \leq |P|, |\mathcal{L}| \leq Uq, \quad d \leq d(v) \leq H. \quad (25)$$

The size lower bounds follow by dividing the surviving incidence count by the maximum degree H .

The pruning also controls the intersection of P with *every* projective line, including lines not in $\mathcal{L}$. We will need this stronger form of control when discarding points on a few lines chosen later. Fix a projective line ℓ . Each point of $P \cap \ell$ is incident with at least $d - 1$ lines of $\mathcal{L}$ different from ℓ . These lines are distinct for different points of $P \cap \ell$. Since $d \rightarrow \infty$, for large q we obtain

$$|P \cap \ell| \leq \frac{|\mathcal{L}|}{d-1} \leq 16U^3q^{1/2} = H \quad \text{for every projective line } \ell. \quad (26)$$

Four anchors with many common witnesses. We seek four points, three collinear and the fourth off their line, together with many points joined to each of them by lines of $\mathcal{L}$. We obtain the four points by first counting such configurations around each possible common point and then averaging.

For $x \in P$, let $N(x)$ consist of the points in $P \setminus \{x\}$ joined to x by a line of $\mathcal{L}$. Distinct lines through x have disjoint point sets away from x , so

$$|N(x)| \geq d(d-1) \geq \frac{q}{128U^4} \quad (27)$$

for large q . Put $t_\ell = |N(x) \cap \ell|$. Summing degrees of the points of $N(x)$ gives

$$\sum_{\ell \in \mathcal{L}} t_\ell \geq d|N(x)| \geq \frac{q^{3/2}}{1024U^6}.$$

Lines with $t_\ell \leq 2$ contribute at most $2Uq$. Because $U = q^{o(1)}$, for sufficiently large q this is at most half of the displayed lower bound. Consequently

$$\sum_{\ell: t_\ell \geq 3} t_\ell \geq \frac{q^{3/2}}{2048U^6}.$$

For every integer $t \geq 3$, one has $t(t-1)(t-2) \geq (2/9)t^3$. Hölder's inequality therefore shows that the number of ordered triples of distinct points of $N(x)$ lying on a line of $\mathcal{L}$ is at least

$$\begin{aligned} \sum_{\ell \in \mathcal{L}} t_\ell(t_\ell-1)(t_\ell-2) &\geq \frac{2}{9|\mathcal{L}|^2} \left(\sum_{\ell: t_\ell \geq 3} t_\ell \right)^3 \\ &\geq \frac{q^{5/2}}{9 \cdot 2^{32}U^{20}}. \end{aligned} \quad (28)$$

By (26) and (27), each such triple admits at least

$$\frac{q}{128U^4} - H \geq \frac{q}{256U^4}$$

choices of a fourth point in $N(x)$ outside its line, once q is large enough. Summing over x and using (25), we obtain at least

$$\frac{q^{9/2}}{9 \cdot 2^{45}U^{28}} \quad (29)$$

ordered choices (x, p_1, p_2, p_3, p_4) with p_1, p_2, p_3 distinct and collinear on a line of $\mathcal{L}$, with p_4 outside that line, and with all four p_i in $N(x)$.

We now reverse the count to fix the four anchor points while retaining many choices of the common point x . The number of possible ordered quadruples (p_1, p_2, p_3, p_4) in this count is at most

$$|\mathcal{L}|H^3|P| \leq 2^{12}U^{11}q^{7/2}. \quad (30)$$

Some quadruple thus has at least

$$c_0 \frac{q}{U^{39}} \quad \text{witnesses } x, \quad c_0 = \frac{1}{9 \cdot 2^{57}}. \quad (31)$$

Discard the witnesses on the common line of p_1, p_2, p_3 and on the joins of p_1, p_4 and of p_2, p_4 . The cap (26) discards at most $3H$ witnesses. Since $U = q^{o(1)}$, this is eventually less than $c_0q/(2U^{39})$. At least that many witnesses remain. Their joins to the four anchors will supply the graph and the four small sets required by Lemma 2.3.

The four pencils as coordinates. Choose projective coordinates in which

$$p_1 = (1 : 0 : 0), \quad p_2 = (0 : 1 : 0), \quad p_3 = (1 : 1 : 0), \quad p_4 = (0 : 0 : 1).$$

This is possible over $\mathbb{F}_q$: representatives of p_1, p_2 span their line, the representative of the distinct third point has both coefficients nonzero and permits rescaling of the first two, and a representative of p_4 completes a basis. The discarded lines are precisely the coordinate line at infinity and the two coordinate axes through p_4 . Hence every remaining witness has coordinates $(a : b : 1)$ with $a, b \neq 0$.

Let G be the set of these pairs (a, b) , and let A, B be its coordinate projections. Define D and J from G as in Lemma 2.3. The chosen coordinates make the roles of the four pencils explicit: for a witness $(a : b : 1)$, the joins to p_2, p_1, p_3, p_4 , respectively, have equations

$$X = aZ, \quad Y = bZ, \quad X - Y = (a - b)Z, \quad X = (a/b)Y.$$

Each join is a member of $\mathcal{L}$, by the witness property. Each anchor belongs to P and has degree at most H . Distinct values of the corresponding parameter give distinct lines in its pencil. Thus

$$|A|, |B|, |D|, |J| \leq H, \quad |G| \geq \frac{c_0 q}{2U^{39}}.$$

With

$$n = q^{1/2}, \quad V = \frac{2}{c_0} U^{39}, \tag{32}$$

we have $V \geq 2$, $V = q^{o(1)}$, and all the hypotheses (8) hold.

The incidence configuration now meets the extraction hypotheses. Lemma 2.3 gives $S \subseteq \mathbb{F}_q^*$ satisfying

$$\frac{3q^{1/2}}{8V^2} \leq |S| \leq Vq^{1/2}, \quad |S - S|, |S/S| \leq 2^{20} V^{22} q^{1/2}.$$

In particular $|S| = q^{1/2+o(1)}$, and, on setting $K = 2^{23} V^{24}$, we have $K \geq 2$, $K = q^{o(1)}$, and $|S - S|, |S/S| \leq K|S|$. This contradicts Lemma 2.5.

We have excluded every sequence of counterexamples of the form (22). The negation of the theorem would have produced such a sequence, so some absolute $\kappa \in (0, 1/10)$ and Q satisfy the stated conclusion. $\square$

3 Line covers after random deletions

We regard the affine plane $\mathbb{F}_q^2$ as a subset of its projective completion $\mathbb{P}^2(\mathbb{F}_q)$. Thus a pencil with centre at infinity is a parallel class of affine lines. The purpose of this section is to show that a set of at most q lines covering almost all survivors of a suitable random deletion process must contain almost a full pencil. The deletions within a single trial may be dependent; independence is required only between trials.

Classical covering results for finite planes are closely related by duality to blocking sets. Jamison and Brouwer–Schrijver established the affine blocking-set lower bound $2q - 1$ [23, 10], and Blokhuis, Brouwer, and Szőnyi studied partial projective covers and bounds depending on their uncovered points [8, Section 1.1, Proposition 1.5]. Here the covered set instead consists of the positive-density survivors of structured random deletions. We prove the pencil statement needed in this setting directly.

There are two different concentration tasks. The total number of surviving points and the number on a fixed line have bounded changes under one trial; for these we use the elementary

estimates in [27, Lemmas A.2–A.3]. The more delicate task is to retain many intersections of a short list of lines, simultaneously for every such list. A deletion trial rarely hits this small set, and we must use that rarity in the variance estimate. We prove the needed Bernstein-type martingale bound here, following the method of Freedman [16, Theorem (1.6) and Proposition (2.1)].

Lemma 3.1 (Martingale concentration). *Let Z_i , $1 \leq i \leq n$, be martingale differences for a filtration $(\mathcal{A}_i)_{i=0}^n$. Suppose that $M, V > 0$ and deterministic numbers $v_i \geq 0$ satisfy, almost surely,*

$$|Z_i| \leq M, \quad \mathbb{E}(Z_i^2 \mid \mathcal{A}_{i-1}) \leq v_i, \quad \sum_{i=1}^n v_i \leq V.$$

Then, for every $u > 0$,

$$\Pr \left(\left| \sum_{i=1}^n Z_i \right| \geq u \right) \leq 2 \exp \left[-\frac{1}{4} \min \left\{ \frac{u^2}{V}, \frac{u}{M} \right\} \right]. \quad (33)$$

Proof. For $|z| \leq 1$ we have $e^z \leq 1 + z + z^2$. Consequently, if $|\theta| \leq M^{-1}$, the conditional mean-zero property gives

$$\mathbb{E}(e^{\theta Z_i} \mid \mathcal{A}_{i-1}) \leq 1 + \theta^2 v_i \leq e^{\theta^2 v_i}.$$

Iteration and Markov's inequality bound the upper tail by $\exp(\theta^2 V - \theta u)$ for $0 < \theta \leq M^{-1}$. Take $\theta = \min\{u/(2V), 1/M\}$. If $u \leq 2V/M$, the exponent is $-u^2/(4V)$; otherwise it is at most $-u/(2M)$. Applying the same argument to $-Z_i$ proves (33). $\square$

Lemma 3.2 (Line covers after random deletions). *Let $q \rightarrow \infty$ through prime powers for which every proper subfield of $\mathbb{F}_q$ has order at most $q^{1/3}$. Fix a positive integer D and constants $\delta > 0$ and $0 < \rho < 1$. For each q , let $\mathcal{F}$ be a family of at most Dq^D subsets of $\mathbb{F}_q^2$, called shapes, such that*

1. *every $F \in \mathcal{F}$ has at most Dq points;*
2. *for every $F \in \mathcal{F}$ and affine line ℓ , either $F = \ell$ or $|F \cap \ell| \leq D$.*

There are at most Dq independent trials, each producing a random list of at most D shapes from $\mathcal{F}$. Delete every point on any listed shape, and denote the surviving set by X . Assume that

1. *for each point x and each trial, the probability that this trial deletes x is at most D/q ;*
2. $\Pr(x \in X) = \delta_q \geq \delta$ *for every $x \in \mathbb{F}_q^2$;*
3. *for each affine line ℓ , the sum over trials of the probabilities that the trial lists the exact shape ℓ is at most D/q .*

Write X^ℓ for the surviving set when all exact occurrences of the shape ℓ are ignored, with every other listed shape still deleted. With probability tending to one, both of the following statements hold:

1. $|X^\ell \cap \ell| \geq \delta q/2$ *for every affine line ℓ ;*
2. *every set of at most q affine lines covering all but at most $2q$ points of X contains at least $(1 - \rho)q$ lines through a common projective point.*

Proof. We first establish a simultaneous lower bound on surviving cross-intersections for every admissible pair of short line lists, ignoring any deletion shape equal to one of those lines. After fixing such a realization, we show that any cover lacking an almost full pencil contains an admissible

pair with too few surviving cross-intersections. This second step samples only within the fixed cover, with separate arguments according to its maximum concurrency.

Fix the absolute constant $\kappa \in (0, 1/10)$ of Theorem 2.1, and put $\epsilon = \kappa/10$. All constants below may depend on the fixed parameters, but not on q , the particular deletion distributions, or a subsequently chosen cover.

Uniform point and line counts. Expose the independent trials one at a time. Changing one trial changes $|X|$ by at most $2D^2q$, because the old and new lists contain at most $2D$ shapes in total and each has at most Dq points. The exposure observation in [27, Lemma A.3] therefore bounds the absolute martingale increment by $L = 2D^2q$. There are $T \leq Dq$ trials, and $\mathbb{E}|X| = \delta_q q^2$. The fixed-time concentration estimate in [27, Lemma A.2], with $u = q^{3/2+\epsilon}$, gives

$$\Pr(|X| - \delta_q q^2 \geq q^{3/2+\epsilon}) \leq 2 \exp\left(-\frac{q^{2\epsilon}}{32D^5}\right).$$

If there are no trials, the count is deterministic and this bound holds as well. Equivalently, we may add a dummy trial of zero change.

For the line count $|X^\ell \cap \ell|$, the rule ignoring exact copies of ℓ is applied separately to each trial. Changing one trial then changes the count by at most $2D^2$: each of its at most $2D$ old or new shapes meets ℓ in at most D points. With $L = 2D^2$ and $u = q^{1/2+\epsilon}$, the same two elementary estimates and $T \leq Dq$ give

$$\Pr(|X^\ell \cap \ell| - \mathbb{E}|X^\ell \cap \ell| \geq q^{1/2+\epsilon}) \leq 2 \exp\left(-\frac{q^{2\epsilon}}{32D^5}\right).$$

Moreover, a point restored by ignoring ℓ can be restored only if some trial lists ℓ exactly. The probability of this event is at most D/q , by the third deletion hypothesis. Therefore

$$\delta_q q \leq \mathbb{E}|X^\ell \cap \ell| \leq \delta_q q + D.$$

There are $q(q+1)$ affine lines. The superpolynomial tails and a union bound consequently show that, with probability tending to one,

$$\begin{aligned} |X| &= \delta_q q^2 + O_D(q^{3/2+\epsilon}), \\ |X^\ell \cap \ell| &= \delta_q q + O_D(q^{1/2+\epsilon}) \quad \text{for every affine line } \ell. \end{aligned} \tag{34}$$

The lower line estimate proves the first conclusion of the lemma. For the pencil conclusion we will also need to bound overlaps in an alleged cover. The same estimates give the uniform upper bound

$$|X \cap \ell| \leq \delta_q q + O_D(q^{1/2+\epsilon}), \tag{35}$$

because $X \subseteq X^\ell$.

A simultaneous event for small line samples. The counts just proved control a single line but do not yet restrict how a cover's lines meet. We now obtain a stronger test on short lists of lines: provided their intersections are distinct and not concentrated on a deletion shape, many of those intersections survive. The list length will be logarithmic in q , so that the variance-sensitive tail can pay for a union bound over all lists.

Set

$$h = 10(D+1), \quad H_0 = 2Dh, \quad t = \lceil A \log q \rceil,$$

where the fixed constant A will be chosen below. Consider any ordered tuple of $2t$ distinct affine lines, separated into two lists of length t . Call the tuple admissible if its cross-list intersections form a set P_0 of t^2 distinct finite points, and

$$|F \cap P_0| \leq H_0 \quad \text{for every } F \in \mathcal{F} \text{ which is not one of the tuple lines.} \quad (36)$$

For this tuple, ignore deletions by every shape equal to a tuple line, and let Y_0 count the survivors in P_0 after the remaining deletions. We claim that, with probability tending to one,

$$Y_0 \geq \frac{\delta t^2}{2} \quad \text{for every admissible tuple.} \quad (37)$$

To prove the claim, fix a tuple in advance. Omitting shapes is a deterministic operation on each trial, so the trials remain independent. Omission can only restore points, and hence $\mathbb{E}Y_0 \geq \delta t^2$. By (36), each trial's contribution to the deletion set within P_0 has size at most $C_0 = DH_0$. The probability that this contribution is nonempty is at most Dt^2/q , by a union bound over P_0 and the per-point deletion hypothesis.

For completeness, the rarity of a nonempty contribution also controls the conditional variance of the exposure martingale. Fix the past, and let $g_i(w)$ be the conditional expected final value of Y_0 when the current trial has outcome w . Let $g_i(\emptyset)$ denote the hypothetical value obtained by making this trial delete nothing on P_0 . Independence leaves the law of the future trials unchanged, so

$$|g_i(w) - g_i(\emptyset)| \leq C_0,$$

with equality to zero whenever the trial's contribution is empty. The current trial also retains its original law conditional on the past. Thus the martingale increment has absolute value at most $2C_0$, and its conditional variance is at most

$$\mathbb{E}((g_i(w) - g_i(\emptyset))^2 \mid \text{past}) \leq C_0^2 \frac{Dt^2}{q}.$$

The sum of conditional variances is at most $D^2 C_0^2 t^2$, since there are at most Dq trials. Applying Lemma 3.1 at deviation $\delta t^2/2$ yields

$$\Pr \left(Y_0 < \frac{\delta t^2}{2} \right) \leq 2e^{-c_0 t^2}, \quad c_0 = \min \left\{ \frac{\delta^2}{16D^2 C_0^2}, \frac{\delta}{16C_0} \right\} > 0. \quad (38)$$

In particular, c_0 is independent of A .

There are at most $(q(q+1))^{2t}$ ordered tuples. The logarithm of the union bound in (38) is

$$-c_0 t^2 + 4t \log q + O(t/q) + O(1) = (-c_0 A^2 + 4A + o(1))(\log q)^2.$$

Choose, for example, any fixed $A > 5/c_0$. This proves (37) simultaneously for all admissible tuples.

Consequences of an alleged cover. The random part of the argument is complete. We now keep the deletion outcome fixed and show deterministically that the simultaneous events exclude any cover without an almost full pencil. Random sampling below is only a way to find a short witness inside a fixed cover.

Fix a realization satisfying both (34) and (37). Suppose that the second conclusion of the lemma fails. From an offending cover remove every line which actually occurred as a listed deletion shape.

Such a line contains no point of X , so the remaining line set L still covers all but at most $2q$ points of X . Put $m = |L|$, and for any projective point p write $n_p = |\{\ell \in L : p \in \ell\}|$. Define

$$E_L = \sum_{x \in X} (n_x - 1)_+, \quad (a)_+ = \max\{a, 0\}.$$

The count of covered survivors and (35) give

$$m(\delta_q q + O_D(q^{1/2+\epsilon})) \geq \delta_q q^2 - O_D(q^{3/2+\epsilon}) - 2q.$$

Since $\delta_q \geq \delta$ and $m \leq q$, this implies

$$m = q - O_{D,\delta}(q^{1/2+\epsilon}), \quad E_L = O_D(q^{3/2+\epsilon}). \quad (39)$$

For the second estimate, subtract the number of covered survivors from $\sum_{\ell \in L} |X \cap \ell|$ and use $m \leq q$. Finally, put

$$R = \max_{p \in \mathbb{P}^2(\mathbb{F}_q)} n_p.$$

The failure of the pencil conclusion implies $R \leq (1 - \rho)q$. We will find an admissible tuple of lines from L with fewer than $\delta t^2/2$ survivors in its cross-intersection set.

Low concurrency: $R \leq q^{1/2+\kappa/4}$. Let $T = q^{1/2-\kappa/3}$. The number of ordered pairs of distinct lines of L meeting at a point $x \in X$ is $\sum_{x \in X} n_x(n_x - 1)$. Points with $n_x < T$ contribute at most

$$TE_L = O(q^{2-7\kappa/30}). \quad (40)$$

The set $P = \{x \in X : n_x \geq T\}$ has size

$$|P| \leq \frac{E_L}{T-1} = O(q^{1+13\kappa/30}) \leq q^{1+\kappa}$$

for sufficiently large q . Theorem 2.1, applied to P and L , therefore gives $\sum_{x \in P} n_x \leq q^{3/2-\kappa}$. The contribution of these points to the ordered-pair count is at most

$$R \sum_{x \in P} n_x \leq q^{2-3\kappa/4}. \quad (41)$$

Together, (40) and (41) show that the fraction of ordered distinct line pairs meeting in X is

$$O(q^{-7\kappa/30} + q^{-3\kappa/4}) = o(1). \quad (42)$$

Sample an ordered tuple of $2t$ distinct lines uniformly from L . The expected number of its cross pairs meeting in X is t^2 times the fraction in (42). Markov's inequality shows that this number is less than $\delta t^2/2$ with probability tending to one.

We next check the two admissibility conditions. A given sampled pair is parallel with probability at most $R/(m-1)$. Once two distinct lines have been sampled, at most R population lines pass through their projective intersection, so the probability of concurrency for any specified three sampled positions is at most $R/(m-2)$. The probability of any parallel pair or concurrent triple in the tuple is consequently

$$O\left(\frac{(t^2 + t^3)R}{m}\right) = O((\log q)^3 q^{-1/2+\kappa/4}) = o(1). \quad (43)$$

If neither event occurs, every cross intersection is finite and the cross intersections are distinct: equality of two different cross intersections would make at least three of the tuple lines concurrent.

To verify (36), fix $F \in \mathcal{F}$. Consider the event that F is not a tuple line but contains more than H_0 distinct finite cross intersections. Each tuple line then contains at most D points of F . Greedily choose a cross intersection on F and its two representing positions, and discard all cross intersections on either of those lines. Each step discards at most $2D$ points. Since $H_0 = 2Dh$, this produces h cross pairs whose $2h$ positions are distinct.

Among the population L , there are at most mDR ordered distinct line pairs, neither line equal to F , whose intersection lies in F . Indeed, the first line meets F in at most D points, and at each such point there are at most R choices of the second line. There are at most t^{2h} ordered choices of the h cross pairs of positions. For each choice, the sampled lines at those $2h$ distinct positions are uniform among $(m)_{2h}$ assignments, where $(z)_a = z(z-1)\cdots(z-a+1)$. The probability of the event for this fixed shape is therefore at most

$$\frac{t^{2h}(mDR)^h}{(m)_{2h}} = O_{D,h} \left(\left(\frac{t^2 R}{m} \right)^h \right). \quad (44)$$

The numerator may count assignments with repeated population lines; this only enlarges the upper bound. Summing (44) over all shapes gives

$$O((\log q)^{2h} q^{D-h(1/2-\kappa/4)}) = o(1),$$

because $\kappa < 1/10$ and $h = 10(D+1)$ imply

$$D - h(1/2 - \kappa/4) < D - \frac{19}{4}(D+1) < 0.$$

Thus, with probability tending to one, the sampled tuple is admissible and has fewer than $\delta t^2/2$ cross pairs meeting in X .

High concurrency: $R > q^{1/2+\kappa/4}$. Choose a projective point p with $n_p = R$, and take all R lines of L through p as the pencil. Its complement has size

$$N = m - R \geq \rho q/2$$

for sufficiently large q , by (39); no complement line passes through p . Sample the first list of t lines uniformly without replacement from the pencil, and independently sample the second list in the same way from the complement.

Any cross intersection lies off p . At a cross intersection $x \in X$, exactly one pencil line passes through x , and the number of cross pairs meeting there is the number of complement lines through x , namely $n_x - 1$. Thus at most E_L of all RN population cross pairs meet in X . Their fraction is at most

$$\frac{E_L}{RN} = O(q^{\epsilon-\kappa/4}) = O(q^{-3\kappa/20}) = o(1). \quad (45)$$

As before, expectation and Markov's inequality show that fewer than $\delta t^2/2$ sampled cross pairs meet in X , with probability tending to one.

For a fixed complement line, at most one pencil line meets it at infinity when p is finite. When p is at infinity, the pencil lines have direction p and every complement line has a different direction, so all cross intersections are finite. The probability of any infinite cross intersection is therefore at most t^2/R . Two equal cross intersections must use the same pencil line, since distinct pencil lines

meet only at p , which lies on no complement line. Given two complement lines, their projective intersection determines at most one affine pencil line, namely its join with p . If this join is the line at infinity, there is no affine candidate. Conditioning on the second list and taking a union bound over its pairs shows that the probability of a collision is $O(t^3/R)$. The total probability of a failure of finiteness or distinctness is therefore

$$O\left(\frac{t^2 + t^3}{R}\right) = O((\log q)^3 q^{-1/2-\kappa/4}) = o(1). \quad (46)$$

Fix $F \in \mathcal{F}$, and consider the event that F is not a tuple line but contains more than H_0 distinct finite cross intersections. The same greedy argument gives h cross pairs with disjoint positions. There are at most DN population cross pairs, with neither line equal to F , whose intersection lies in F : a complement line distinct from F meets F in at most D points, all off p , and each such point determines at most one pencil line. The two lists are sampled independently, so the probability of this occupancy violation for a fixed shape is at most

$$\frac{t^{2h}(DN)^h}{(R)_h(N)_h} = O_{D,h}\left(\left(\frac{t^2}{R}\right)^h\right). \quad (47)$$

After summing over $\mathcal{F}$, the bound becomes

$$O((\log q)^{2h} q^{D-h(1/2+\kappa/4)}) = o(1),$$

since

$$D - h(1/2 + \kappa/4) < D - 5(D+1) < 0.$$

The sampled tuple is consequently admissible and has fewer than $\delta t^2/2$ cross pairs meeting in X , with probability tending to one in this regime as well.

The contradiction and the order of choices. All sampling estimates above are uniform over line sets satisfying (39) and the appropriate bound on R . In either regime, for every sufficiently large q there is therefore an admissible tuple from the fixed line set L with

$$|X \cap P_0| < \delta t^2/2.$$

No line of L actually occurred as a listed deletion shape. Hence ignoring exact occurrences of its tuple lines changes none of the deletions in the fixed realization, and the tuple's count Y_0 is exactly $|X \cap P_0|$. This contradicts (37).

In particular, the cover and the tuple are allowed to depend on the realized deletions: the event (37) was established simultaneously for every admissible tuple before fixing the realization. Sampling from L is used only to prove the existence of a tuple inside that fixed realization. No concentration estimate is conditioned on the choice of the cover. The contradiction proves the second conclusion, and finishes the proof. $\square$

4 Selecting compatible translates

The deletion estimates require independent random choices, while intersection of the exceptional edges requires geometric compatibility. We separate these tasks by sampling several independent translates of each bounded configuration and then selecting one from each pool. The theorem below

supplies the selection step. Its input consists of finite point sets W_i and sets D_i of forbidden join directions; its output avoids every forbidden join between different selected sets. Because a direction is a linear subspace containing zero, the condition also rules out a common point between two selected configurations whenever at least one of their direction sets is nonempty.

Theorem 4.1 (Compatible translates from independent pools). *For every positive integer D there is a positive integer $K = K(D)$ with the following property. Let q tend to infinity through prime powers. For each q , let $n \leq Dq$, and for $1 \leq i \leq n$ let $W_i \subseteq \mathbb{F}_q^2$ and let D_i be a set of directions, with*

$$|W_i| \leq D, \quad |D_i| \leq D.$$

Assume that each direction belongs to D_i for at most D indices i . For distinct indices i, j , declare the translates $b + W_i$ and $b' + W_j$ to conflict if there are $u \in W_i$, $v \in W_j$, and $d \in D_i \cup D_j$ such that

$$(b + u) - (b' + v) \in d.$$

For each i , draw K independent uniform shifts $b_{i,1}, \dots, b_{i,K} \in \mathbb{F}_q^2$, independently for all indices. With probability tending to one, one can choose $k_i \in \{1, \dots, K\}$ for every i so that the translates $b_{i,k_i} + W_i$ have no conflicts. The convergence is uniform over all data satisfying the stated bounds.

To prove the theorem, make each candidate a vertex of a graph and join candidates from different pools when they conflict. We seek an independent set containing one vertex from each pool, called an independent transversal. Its nonexistence has a small domination certificate. We first prove the form of that assertion we need, then show that the independent shifts make every such certificate unlikely. The certificate argument is a weaker form of Haxell's independent-transversal domination theorem [19], whose method originates in [18]. We include its proof. Haxell's theorem uses total domination, which also requires neighbors for the vertices of the dominating set. Here we use the following weaker convention. For a graph G and a set S of its vertices, we say that $T \subseteq S$ *dominates* S if every vertex of $S \setminus T$ has a neighbor in T . No condition is imposed on the vertices of T themselves.

Lemma 4.2 (A domination certificate). *Let a finite simple graph have a vertex partition into nonempty parts $A_1, \dots, A_n$. If there is no independent set containing one vertex from each part, then there are a nonempty index set I and a set*

$$T \subseteq \bigcup_{i \in I} A_i, \quad |T| \leq 2(|I| - 1),$$

such that T dominates $\bigcup_{i \in I} A_i$.

Proof. We grow a set of vertices with small size compared with the number of parts it reaches. Whenever it does not yet dominate those parts, we use an undominated vertex to reach another part. A minimality condition on the transversals prevents this growth from stopping without the required domination certificate.

Choose an inclusion-minimal nonempty subfamily of parts with no independent transversal, and distinguish one part A_* of this subfamily. Let $\mathcal{U}$ be the family of independent transversals of all its other parts. Minimality makes $\mathcal{U}$ nonempty. The deficient subfamily has at least two parts, since every part is nonempty.

Choose any $x_1 \in A_*$. Among $U \in \mathcal{U}$, minimize $|N(x_1) \cap U|$, where $N(x)$ denotes the neighborhood of x in G . Fix a minimizer and write $Y_1 = N(x_1) \cap U$. This set is nonempty: otherwise adjoining x_1 would give an independent transversal of the deficient subfamily.

Inductively, suppose that vertices $x_1, \dots, x_j$ and sets $Y_1, \dots, Y_j$ have been fixed with the following properties:

1. The sets Y_a are nonempty and pairwise disjoint, and some $U \in \mathcal{U}$ jointly realizes all the equalities

$$N(x_a) \cap U = Y_a \quad (1 \leq a \leq j).$$

2. At stage a , the size $|Y_a|$ was minimized over all $U \in \mathcal{U}$ satisfying the already fixed equalities for indices smaller than a .

3. Each x_a lies either in A_* or in a part represented by a vertex of some earlier Y_b , with $b < a$.

Let $Y = \bigcup_{a=1}^j Y_a$, let I consist of the distinguished part and all parts represented in Y , and put

$$T = \{x_1, \dots, x_j\} \cup Y.$$

By joint realization, the vertices of Y occupy distinct parts, none of them A_* . Thus $|I| = |Y| + 1$, while nonemptiness of the Y_a gives $j \leq |Y|$. The third invariant shows that T lies in the union of the parts indexed by I , and

$$|T| \leq j + |Y| \leq 2|Y| = 2(|I| - 1).$$

If T dominates this union, we have the required certificate.

Otherwise, choose a vertex x_{j+1} in that union, outside T , with no neighbor in T . Among the transversals realizing all the current equalities, choose U minimizing $|N(x_{j+1}) \cap U|$, and set

$$Y_{j+1} = N(x_{j+1}) \cap U.$$

This set avoids all the previous Y_a , because x_{j+1} has no neighbor in T . We claim it is nonempty.

If it were empty and $x_{j+1} \in A_*$, then $U \cup \{x_{j+1}\}$ would be an independent transversal of the deficient subfamily. Otherwise, the part of x_{j+1} contains a unique vertex y of Y , say $y \in Y_b$. The transversal U uses y in that part, so $x_{j+1} \notin U$. Replace y by x_{j+1} to form

$$U' = (U \setminus \{y\}) \cup \{x_{j+1}\}.$$

Since $N(x_{j+1}) \cap U = \emptyset$, this is again an independent transversal of the nondistinguished parts. For every $a < b$, the vertex y does not belong to Y_a , and the new vertex is not adjacent to x_a . Consequently the exact earlier equalities are preserved:

$$N(x_a) \cap U' = Y_a \quad (a < b).$$

For $a = b$, the replacement removes the neighbor y and adds no neighbor, so

$$|N(x_b) \cap U'| = |Y_b| - 1.$$

This contradicts the minimum imposed at stage b . Notice that the contradiction uses only the equalities preceding stage b ; preservation of later equalities is not required.

Thus Y_{j+1} is nonempty and all the induction invariants continue to hold. Each continuation adds vertices in previously unrepresented parts to Y , whereas all of Y lies in one transversal of the fixed nondistinguished subfamily. There can be only finitely many continuations. The process must therefore stop with a dominating set T , proving the lemma. $\square$

Proof of Theorem 4.1. The case $n = 0$ is immediate, so suppose $n \geq 1$. Fix D and, for the moment, a constant integer $K \geq 6$. We will choose K sufficiently large in terms of D at the end.

Form a graph whose vertices are the candidate positions (i, k) , partitioned into the n pools. Put an edge between candidates in different pools precisely when their translates conflict. Candidate

positions are distinct vertices even when their sampled shifts happen to coincide. An independent transversal in this graph is exactly a selection as in the theorem.

By Lemma 4.2, it suffices to show that, with probability tending to one, there are no nonempty index sets I , with $|I| = m$, and sets T of at most $2m$ candidate positions in those pools that dominate their union. We use the slightly weaker bound $2m$ to simplify the counting. Throughout the probability estimates, I and T are fixed sets of indices before any shifts are exposed.

The proof splits according to m , the number of pools in the certificate. When m is small, the forbidden lines associated with T occupy only a small part of each candidate's possible shifts. When m is proportional to q , that direct bound is insufficient. Instead, we expose T in a bounded number of chunks and prove that a positive average proportion of shifts remains feasible. Crucially, this proportion will depend only on D , so we may subsequently choose K large enough that the unused candidates defeat all certificates.

Forbidden equations and counting possible certificates. For every direction d in use, fix a nonzero linear functional $\pi_d : \mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ whose kernel is d . After exposing the shifts of the candidates $e \in T$, call a shift $b \in \mathbb{F}_q^2$ *feasible for target i* if, for every such e , with source-pool index j , it avoids every equation

$$\pi_d(b) = \pi_d(b_e) + \pi_d(v - u), \quad u \in W_i, \quad v \in W_j, \quad d \in D_i \cup D_j. \quad (48)$$

We impose these prohibitions even when $i = j$. This only makes feasibility more restrictive. A feasible candidate outside T has no neighbor in T , so its existence prevents domination.

Set

$$C = 2D^3, \quad c_1 = \frac{1}{4C}, \quad \alpha = \frac{1}{2C}. \quad (49)$$

For one target i and one exposed candidate e , there are at most C equations in (48). This remains true if the two lists D_i and D_j are counted separately, retaining duplicate directions. Each equation excludes one affine line, containing exactly q shifts.

The number of choices of I is at most

$$\binom{n}{m} \leq \left(\frac{eDq}{m} \right)^m.$$

There is an absolute constant $C_* > 0$ such that, for $K \geq 6$, the number of candidate-index sets T in these pools with $|T| \leq 2m$ is at most

$$\sum_{t=0}^{2m} \binom{Km}{t} \leq (C_* K^2)^m. \quad (50)$$

For example, the usual bound $\sum_{t=0}^r \binom{N}{t} \leq (eN/r)^r$ for $1 \leq r \leq N/2$ applies with $N = Km$ and $r = 2m$.

The range $1 \leq m \leq c_1 q$. Condition on all the exposed shifts in T . A fresh uniform shift for any target $i \in I$ is infeasible with probability at most

$$\frac{C|T|}{q} \leq \frac{2Cm}{q} \leq \frac{1}{2}.$$

At least $(K - 2)m$ candidate positions in the target pools remain unexposed. Their shifts are conditionally independent and uniform. Thus the probability that T dominates, for this fixed I, T , is at most

$$\left(\frac{2Cm}{q}\right)^{(K-2)m}. \quad (51)$$

Combining this with the counts of I and T , the probability of a dominating certificate of size parameter m is at most

$$\left[2CeDC_*K^2\left(\frac{2Cm}{q}\right)^{K-3}\right]^m. \quad (52)$$

Choose K large enough that

$$2CeDC_*K^2 2^{-(K-3)} \leq \frac{1}{2}. \quad (53)$$

Then the bracket in (52) is at most $1/2$ throughout this range, and tends to zero for every fixed positive m . Summing over m gives $o(1)$: for a fixed cutoff, the finite initial sum tends to zero, while the remaining terms are bounded by a geometric tail.

A feasible-density estimate for $m > c_1q$. We prove that there is a constant $\delta_1 > 0$, depending only on D and not on K , such that for every fixed I, T in this range,

$$\Pr\left[\sum_{i \in I} |U_i^{\text{fin}}| < \delta_1 m q^2\right] \leq \exp(-\Omega_D(q \log q)). \quad (54)$$

Here U_i^{fin} is the set of shifts feasible for target i after all candidates in T have been exposed. The estimate is uniform in I, T and the configuration data. The lower cutoff on q may depend on the fixed value of K .

Expose the candidate positions of T in a fixed order, divided into consecutive chunks of size at most $\lfloor \alpha q \rfloor$, with all but the last having that size. For sufficiently large q , $\lfloor \alpha q \rfloor \geq \alpha q/2$. Since $|T| \leq 2m \leq 2Dq$, the number of chunks is at most

$$J_0 = \left\lceil \frac{4D}{\alpha} + 1 \right\rceil. \quad (55)$$

In particular, this bound does not involve K . If T is empty, the desired feasible-density estimate holds deterministically; otherwise there is at least one chunk.

At the start of a chunk, condition on all shifts previously exposed. Let U_i be the set of shifts still feasible for target i , and write

$$M = \sum_{i \in I} |U_i|.$$

For the analysis of this chunk we keep all these sets frozen at their start values. They are now deterministic, while the shifts belonging to the new chunk are independent uniform vectors.

For an entry e of the chunk, with source-pool index j , define Z_e by summing, over all equations (48) associated with $i \in I$ and e , the number of shifts $b \in U_i$ satisfying that equation. Count the D_i and D_j lists separately, including duplicates. A shift deleted more than once is therefore counted more than once, which is harmless for an upper bound on loss. Each line contains at most q shifts, and each of its right-hand sides is uniform in $\mathbb{F}_q$. Consequently,

$$0 \leq Z_e \leq Cmq, \quad \mathbb{E}Z_e \leq \frac{CM}{q}. \quad (56)$$

All expectations and variances in the following chunk calculation are conditional on the fixed past.

The terms whose directions belong to the target. The expectation bound alone does not control the loss in a chunk. We estimate its fluctuations by separating the two direction lists in each forbidden equation. Bounded reuse of target directions handles one list; the affine second moment will handle the source list. Write

$$Z_e = Z_e^{(1)} + Z_e^{(2)},$$

where $Z_e^{(1)}$ uses the list D_i in each target, and $Z_e^{(2)}$ uses the list D_j of the source. Group the terms of $Z_e^{(1)}$ by direction. The group for d is a function only of $\pi_d(b_e)$. For distinct directions d, d' , the functionals $\pi_d, \pi_{d'}$ are linearly independent, so

$$b_e \mapsto (\pi_d(b_e), \pi_{d'}(b_e))$$

is a bijection from $\mathbb{F}_q^2$ to itself. The two coordinates are therefore independent and uniform. The direction groups are pairwise independent, which suffices for adding their variances.

There are at most Dm such groups. Each direction belongs to at most D target direction sets; for each associated target-source pair there are at most D^2 choices of (u, v) . Thus every group is bounded by D^3q , and

$$\text{Var}(Z_e^{(1)}) \leq D^7 m q^2. \quad (57)$$

The terms whose directions belong to the source. For each target i and each direction d , define the variance of its parallel line slices by

$$V_{i,d} = \frac{1}{q} \sum_{\ell \parallel d} \left(|U_i \cap \ell| - \frac{|U_i|}{q} \right)^2, \quad (58)$$

where the sum is over the q affine lines of direction d . These variances obey the exact identity

$$\sum_d V_{i,d} = |U_i| - \frac{|U_i|^2}{q^2} \leq |U_i| \leq q^2, \quad (59)$$

where the sum runs over all $q+1$ directions. This is precisely the normalized affine second-moment identity in [27, Lemma A.1], applied to $U_i \subseteq \mathbb{F}_q^2$. That identity holds for every subset of every finite affine plane; it has no randomness, characteristic, or subfield hypothesis. Thus it applies after conditioning on any history of earlier chunks. This is the reason we can control all source directions even when the feasible sets U_i have complicated shapes.

Every individual summand of $Z_e^{(2)}$ associated with i, d, u, v is a uniformly sampled line-slice count, with its slice index translated by a fixed offset. Its variance is therefore $V_{i,d}$. There are at most D^3m summands, and each pair (i, d) is repeated at most D^2 times. Cauchy–Schwarz applied to their centered sum, without any independence assumption, gives

$$\text{Var}(Z_e^{(2)}) \leq D^5 m A_e, \quad A_e := \sum_{i \in I} \sum_{d \in D_j} V_{i,d}. \quad (60)$$

Each source pool contributes at most K entries to the chunk, and each direction belongs to at most D source direction sets. Consequently, by (59),

$$\sum_{e \text{ in the chunk}} A_e \leq KD \sum_{i \in I} \sum_d V_{i,d} \leq KDm q^2. \quad (61)$$

Call an entry *regular* if $A_e \leq m q^{3/2}$. There are at most $KD\sqrt{q}$ irregular entries. For a regular entry, (57), (60), and $\text{Var}(X+Y) \leq 2\text{Var}(X) + 2\text{Var}(Y)$ give

$$\text{Var}(Z_e) \leq 2D^7 m q^2 + 2D^5 m^2 q^{3/2}. \quad (62)$$

For a fixed constant $\gamma > 0$, Chebyshev's inequality and (56) therefore imply

$$\begin{aligned} \Pr\left[Z_e > \frac{CM}{q} + \gamma mq\right] &\leq \frac{2D^7}{\gamma^2 m} + \frac{2D^5}{\gamma^2 \sqrt{q}} \\ &\leq B_{D,\gamma} q^{-1/2} \quad \text{for regular } e, \end{aligned} \quad (63)$$

where $B_{D,\gamma}$ is independent of K . In the last step we used $m > c_1 q$.

Conditional independence and loss over one chunk. The preceding variance bounds show that each regular entry has a small chance of removing far more than its expected share. We now use independence within the chunk to show that few entries do so, and convert that statement into a deterministic bound on the total remaining feasible set.

The regularity of an entry is determined by the frozen sets U_i and its fixed source-pool index. In particular, it does not depend on its fresh shift b_e . After conditioning on the past, the collection of regular entries is fixed, and their variables Z_e are independent functions of independent shifts. Their common threshold in (63) is also fixed under this conditioning.

A union bound over subsets of regular entries, using the chunk size bound and (63), shows that the probability of at least γq threshold exceedances is at most

$$2^{\lceil \alpha q \rceil} \left(B_{D,\gamma} q^{-1/2} \right)^{\lceil \gamma q \rceil} \leq \exp(-c_{D,\gamma} q \log q) \quad (64)$$

for all sufficiently large q , with $c_{D,\gamma} > 0$ independent of K . If there are fewer than $\lceil \gamma q \rceil$ regular entries, the event in question is empty and the bound remains valid.

For each fixed K , we may also take q sufficiently large that

$$KD\sqrt{q} \leq \gamma q. \quad (65)$$

Except on the event in (64), all but at most $2\gamma q$ entries are regular and below the threshold: the exceptions include all irregular entries. The actual loss of feasible shifts within the chunk is at most $\sum_e Z_e$, because these counts used the larger feasible sets at the start of the chunk. Bounding each exceptional entry by Cmq and all other entries by the threshold gives, for the end-of-chunk value M' , the deterministic inequality

$$\begin{aligned} M' &\geq M - C\alpha M - \alpha\gamma mq^2 - 2C\gamma mq^2 \\ &= (1 - C\alpha)M - (\alpha + 2C)\gamma mq^2 \\ &= \frac{1}{2}M - (\alpha + 2C)\gamma mq^2. \end{aligned} \quad (66)$$

Choose $\gamma > 0$, depending only on D , so small that

$$2(\alpha + 2C)\gamma \leq 2^{-J_0-1}, \quad \delta_1 := 2^{-J_0-1}. \quad (67)$$

At the beginning, before exposing T , every shift is feasible, so $M = mq^2$. If all $J \leq J_0$ chunks satisfy (66), iteration gives

$$\begin{aligned} \frac{M_{\text{fin}}}{mq^2} &\geq 2^{-J} - (\alpha + 2C)\gamma \sum_{a=0}^{J-1} 2^{-a} \\ &\geq 2^{-J_0} - 2(\alpha + 2C)\gamma \geq \delta_1. \end{aligned}$$

The conditional probability bound (64) holds for every possible past. Averaging over the past and taking a union bound over the at most J_0 chunks proves (54). The constants J_0 , γ , and δ_1 depend only on D . The dependence on K in (65) affects only how large q must be, not these constants.

Unused candidates and the large-range union bound. We have established the required positive average feasible density with a failure probability of order $\exp(-\Omega_D(q \log q))$. There are only exponentially many candidate certificates, so this error can be summed over all of them. On the successful event, the unused candidates now supply the further exponential saving needed to rule out domination.

Condition now on all the shifts belonging to T , and suppose the feasible-density event in (54) succeeds. Let t_i be the number of exposed candidate positions from pool i . Every unused candidate from that pool remains independent and uniform, and is feasible with probability $|U_i^{\text{fin}}|/q^2$. The probability that none of the unused candidates in the target pools is feasible is therefore at most

$$\begin{aligned} \prod_{i \in I} \left(1 - \frac{|U_i^{\text{fin}}|}{q^2}\right)^{K-t_i} &\leq \exp\left(-\sum_{i \in I} (K-t_i) \frac{|U_i^{\text{fin}}|}{q^2}\right) \\ &\leq \exp(-(K\delta_1 - 2)m). \end{aligned} \quad (68)$$

To obtain the last inequality, use

$$\sum_{i \in I} \frac{|U_i^{\text{fin}}|}{q^2} \geq \delta_1 m, \quad 0 \leq \frac{|U_i^{\text{fin}}|}{q^2} \leq 1, \quad \sum_{i \in I} t_i = |T| \leq 2m.$$

This argument also covers pools for which every candidate belongs to T ; it does not assume $K - t_i$ is positive in every target pool.

For $m > c_1 q$, the number of pairs I, T is at most A_K^m , where

$$A_K = \frac{eD}{c_1} C_* K^2. \quad (69)$$

For each fixed pair, domination can occur only if the feasible-density estimate fails or all unused candidates are infeasible despite its success. The sum of the first failure probabilities over all m, I, T in the large range is bounded by

$$\exp(O_{D,K}(q)) \exp(-\Omega_D(q \log q)) = o(1).$$

Here the number of choices is at most $\sum_{m \leq Dq} A_K^m = \exp(O_{D,K}(q))$, with K fixed.

Finally choose K sufficiently large, still depending only on D , so that both (53) and

$$A_K \exp(2 - K\delta_1) < \frac{1}{2} \quad (70)$$

hold. Such a choice exists because $\delta_1 > 0$ does not depend on K , whereas A_K grows only quadratically in K . By (68), the sum of the remaining failure probabilities is then at most

$$\sum_{m > c_1 q} (A_K \exp(2 - K\delta_1))^m \leq \sum_{m > c_1 q} 2^{-m} = o(1).$$

Together with the small-range estimate, this excludes all dominating certificates with probability tending to one. Lemma 4.2 therefore supplies an independent transversal, proving the theorem.

The choices have been made in the order

$$D ; \quad C, c_1, \alpha, J_0, \gamma, \delta_1 ; \quad K ; \quad q.$$

In particular, K is fixed before taking q to infinity, and all sufficiently-large- q conditions may depend on that fixed K . Every estimate above is uniform over the configurations and direction sets subject to the bounds in the theorem. $\square$

5 A configuration with two coordinatizations

We now use the deletion lemma to construct the exceptional configuration. We need two properties of the same set: many points on each undeleted line, and rigidity of its small line covers. A change of coordinates lets us obtain both. In the original coordinates we delete one line in every finite nonzero direction. After the change of coordinates, the deleted sets become curves, so the line-survival conclusion of Lemma 3.2 applies to every line. It will force every small mixed cover by lines and individual points to be a parallel class.

Lemma 5.1. *For every sufficiently large prime $s \equiv 2 \pmod{3}$, there is a set $Y \subseteq \mathbb{F}_s^2$ with the following properties. For each $d \in \mathbb{F}_s^*$, there is an affine line ℓ_d of slope d , and*

$$Y = \mathbb{F}_s^2 \setminus \bigcup_{d \in \mathbb{F}_s^*} \ell_d.$$

For each $d \in \mathbb{F}_s^$, every line of slope d other than ℓ_d contains at least five points of Y . The map*

$$f : \mathbb{F}_s^2 \longrightarrow \mathbb{F}_s^2, \quad f(x, y) = (x^3, y),$$

is a bijection, and every affine line contains at least $s/8$ points of $f(Y)$. Moreover, if a affine lines and b individual points cover $f(Y)$, and

$$a + \frac{b}{2} \leq s, \tag{71}$$

then $a = s$, $b = 0$, and the lines form a full parallel class. Consequently every such mixed cover has weighted cost at least s , and no cover of $f(Y)$ by at most s lines contains two nonparallel lines.

Proof. Let s tend to infinity through primes congruent to $2 \pmod{3}$. Since $\gcd(3, s-1) = 1$, cubing permutes $\mathbb{F}_s^*$ and fixes zero. Thus f is bijective. It maps horizontal lines to horizontal lines and vertical lines to vertical lines. We first prove the two survival properties simultaneously with high probability, then use them to classify mixed covers satisfying (71).

Survivors in the original coordinates. Choose the lines ℓ_d , independently for the $s-1$ values of d , uniformly within their parallel classes. Fix a line ℓ of finite nonzero slope d , and temporarily ignore the choice ℓ_d . Each of the other $s-2$ choices deletes a uniformly distributed point of ℓ . Let Z_ℓ count the surviving points on ℓ . Its expectation is

$$\mu := \mathbb{E}Z_\ell = s(1 - 1/s)^{s-2}.$$

Changing one of these choices changes the survivor count by at most two. Apply the independent-trial jump bound and bounded-drift concentration estimate of [27, Lemmas A.2–A.3] to the exposure martingale, with $T = s-2$, $L = 2$, and $u = \mu/2$. The martingale has zero drift, and hence

$$\Pr(Z_\ell < \mu/2) \leq 2 \exp\left(-\frac{\mu^2}{128(s-2)}\right) = \exp(-\Omega(s)).$$

There are fewer than s^2 lines under consideration. A union bound, together with $\mu/2 \geq 5$ for sufficiently large s , shows that all these counts are at least five with probability tending to one. If $\ell \neq \ell_d$, reinstating the deletion by ℓ_d changes no point on ℓ , since the two lines are parallel. Thus every required line in the original coordinates has at least five points of Y .

Survivors and almost-pencil covers in the new coordinates. In the new coordinates, the possible deletion shapes are

$$F_{d,z} = \{(x^3, dx + z) : x \in \mathbb{F}_s\}, \quad d \in \mathbb{F}_s^*, \quad z \in \mathbb{F}_s.$$

There are $s(s-1)$ such shapes, each of size s , and each of the $s-1$ independent trials chooses one of them. Every point is deleted by a specified trial with probability $1/s$, so its overall survival probability is the uniform value

$$\delta_s = (1 - 1/s)^{s-1} \geq \frac{1}{4}$$

for all sufficiently large s .

Each shape meets every affine line in at most three points. For a vertical line this follows from bijectivity of cubing; for a horizontal line it follows from $d \neq 0$. For a line of finite nonzero slope m , the intersection is determined by

$$dx + z = mx^3 + u,$$

a nonzero polynomial equation of degree three. In particular, no shape is itself a line when $s > 5$. All the hypotheses of Lemma 3.2 hold with field size s , a fixed integer constant, for example $D = 3$, survival lower bound $\delta = 1/4$, and $\rho_0 = 1/100$. The exact-line occurrence probabilities are zero, and the subfield restriction is automatic for a prime field. With probability tending to one, every line consequently contains at least $s/8$ points of $f(Y)$, and its almost-cover conclusion holds for $f(Y)$ as well. We now combine that conclusion with the lower bound on every line to replace an almost pencil by a complete one.

From an almost pencil to a full parallel class. Assume these events hold, and consider a cover satisfying (71). Repeated resources can be removed; alternatively, one may keep their original counts, since removal only reduces the budget. The line portion has at most s distinct lines and leaves at most $b \leq 2s$ points of $f(Y)$ uncovered. Lemma 3.2 supplies at least $(1 - \rho_0)s$ distinct lines in this portion through a common projective point p . There are at most $\rho_0 s$ other lines, and

$$b \leq 2(s - a) \leq 2\rho_0 s.$$

If an affine line through p were absent from the cover, all the chosen lines through p would cover at most one of its affine points. The other lines would cover at most $\rho_0 s$ further points, and the individual point resources at most $2\rho_0 s$. Thus at most $1 + 3\rho_0 s$ points of this missing line would be covered. For sufficiently large s , this contradicts

$$\frac{s}{8} > 1 + 3\rho_0 s.$$

Every affine line through p must therefore be present. A finite p requires $s+1$ lines, which is impossible. An infinite p requires all s lines in its parallel class, and then (71) forces $a = s$ and $b = 0$. This also rules out repetitions in a cover attaining the budget.

The events used in the two coordinate systems hold simultaneously with probability tending to one. Consequently every sufficiently large prime $s \equiv 2 \pmod{3}$ admits a successful configuration Y . A mixed cover of cost less than s would satisfy (71) but not its necessary equality conclusion. The two stated consequences follow. $\square$

There are arbitrarily large primes congruent to $2 \pmod{3}$. Indeed, if the list were finite and its product were P , the integer $3P - 1 \equiv 2 \pmod{3}$ would have a prime factor congruent to $2 \pmod{3}$, and none of the primes dividing P could divide it. Fix any sufficiently large prime $s \equiv 2 \pmod{3}$, and choose Y and the omitted lines from Lemma 5.1. These data remain fixed for the rest of the proof.

6 Ambient placement

We will place copies of Y in a larger affine plane, assign each copy the directions in which its labels will change, and delete the ambient lines of those directions that meet it. The placement must meet two requirements. Joins between different copies must avoid their changed directions, while the undeleted points must retain the pencil and line-survival properties of Lemma 3.2. We first specify the copies and direction sets, then prove the deletion estimates for independent pools of translates, and finally select one compatible translate from each pool.

6.1 Fields, copies, and direction sets

Let $n \geq 3$ be odd, and put $q = s^n$. We identify $\mathbb{F}_s$ with the prime subfield of $\mathbb{F}_q$. If E is a subfield, then $|E| = s^a$, where $a = [E : \mathbb{F}_s]$, and multiplication of vector-space bases gives

$$n = a[\mathbb{F}_q : E].$$

For a proper subfield, the index $[\mathbb{F}_q : E]$ is an odd integer larger than one, so it is at least three. Consequently

$$|E| = s^a \leq s^{n/3} = q^{1/3} \quad \text{for every proper subfield } E.$$

Thus Theorem 2.1 and Lemma 3.2 apply along this sequence of ambient fields. In this section all constants may depend on the already fixed s and Y , but not on n .

The odd-degree condition has now supplied the required subfield bound. It remains to assign every ambient direction to at least one copy, with bounded overlap between assignments. Identify the ambient directions with $\mathbb{P}^1(\mathbb{F}_q) = \mathbb{F}_q \cup \{\infty\}$, using slope. Choose a set Λ of representatives of $\mathbb{F}_q^*/\mathbb{F}_s^*$, with $1 \in \Lambda$. There will be one configuration group for each $\lambda \in \Lambda$, together with one further group denoted by $*$. Write

$$\mathcal{I} = \Lambda \sqcup \{*\}, \quad N_q = |\mathcal{I}| = \frac{q-1}{s-1} + 1.$$

Each index $i \in \mathcal{I}$ will specify a pool of translates of one configuration. We call this pool a group and will choose one translate from each group.

For $\lambda \in \Lambda$, define

$$M_\lambda(x, y) = (x, \lambda y), \quad S_\lambda = \lambda \mathbb{F}_s^*, \quad A_\lambda = \{0, \infty\}.$$

Choose distinct $u, v \in \mathbb{F}_s^*$ and define

$$M_* = \begin{pmatrix} 1 & 1 \\ u & v \end{pmatrix}, \quad S_* = (\mathbb{F}_s \cup \{\infty\}) \setminus \{u, v\}, \quad A_* = \{u, v\}.$$

All these linear maps are invertible. For M_* , the image of a finite slope t is

$$\frac{u + vt}{1 + t},$$

where a zero denominator gives the infinite direction. Its images of the horizontal and vertical directions are u and v . The induced projective bijection therefore maps the $s-1$ finite nonzero subfield slopes precisely to S_* . The same statement for M_λ follows from the slope map $t \mapsto \lambda t$.

For each $i \in \mathcal{I}$, put

$$W_i = M_i(\mathbb{F}_s^2).$$

The directions of joins of distinct points of W_i form exactly $S_i \cup A_i$. Each set S_i has size $s - 1$, and A_i consists of the two images of the coordinate axes. The cosets S_λ partition all nonzero finite ambient slopes. The extra set S_* contains 0 and ∞ . Hence every ambient direction belongs to at least one and at most two sets S_i . More precisely, the directions belonging to two sets are

$$S_1 \cap S_* = \mathbb{F}_s^* \setminus \{u, v\}, \quad (72)$$

and the two groups involved are always 1 and $*$.

The sets S_i are the directions whose labels will change, and A_i are the images of the two axis directions preserved by f . For compatibility between copies we use the slightly larger direction sets

$$D_i = \begin{cases} S_i \cup A_i, & i \in \{1, *\}, \\ S_i, & i \notin \{1, *\}. \end{cases}$$

In particular,

$$D_1 = D_* = \mathbb{P}^1(\mathbb{F}_s), \quad (73)$$

whereas the remaining D_i are the nonidentity multiplicative cosets of $\mathbb{F}_s^*$. Every direction occurs in at most two sets D_i . Since $|W_i| = s^2$, $|D_i| \leq s + 1$, and $N_q = O_s(q)$, a fixed constant depending only on s satisfies all size and multiplicity assumptions of Theorem 4.1. In a conflict-free selection, including the axes in both D_1 and D_* ensures that an axis line meeting one of these two selected copies cannot also meet the other. This will enter the weighted cover argument for the two groups sharing a direction in (72).

6.2 Independent candidate pools

Let K be a fixed pool size supplied by Theorem 4.1. For each $i \in \mathcal{I}$, choose independent uniform shifts $b_{i,1}, \dots, b_{i,K} \in \mathbb{F}_q^2$, independently between groups, and write

$$P_{i,k} = b_{i,k} + W_i, \quad Y_{i,k} = b_{i,k} + M_i Y.$$

Associate to this candidate the following set of ambient deletion lines:

$$\mathcal{B}_{i,k} = \{\ell : \text{dir}(\ell) \in S_i, \ell \cap Y_{i,k} \neq \emptyset\}.$$

Lemma 6.1. *For every candidate and every $d \in S_i$, exactly $s - 1$ lines of $\mathcal{B}_{i,k}$ have direction d . Each of these lines contains at least five points of $Y_{i,k}$. Consequently $|\mathcal{B}_{i,k}| = (s - 1)^2$.*

Proof. Undo the translation and apply M_i^{-1} . An ambient line in the given direction meeting $P_{i,k}$ becomes a line through a point of $\mathbb{F}_s^2$ in a finite nonzero subfield direction. Its intersection with $\mathbb{F}_s^2$ is the corresponding subfield line. The s lines in that subfield parallel class extend to distinct parallel ambient lines. Exactly one is the omitted line in the definition of Y ; the other $s - 1$ each contain at least five points of Y , by Lemma 5.1. These are precisely the lines meeting the candidate's retained configuration. $\square$

Each candidate therefore prescribes a bounded list of deletion lines. We next apply the deletion lemma to all these independent lists at once. Besides the almost-pencil and line-survival conclusions, we need one bound on how many groups a line can meet along joins from an external point. All three properties will survive the subsequent selection of one candidate per group.

Lemma 6.2. *There are constants $\delta \in (0, 1)$ and $\rho = \delta/10$, independent of n , with the following properties with probability tending to one as n tends to infinity through odd integers. Let*

$$X_{\text{pool}} = \mathbb{F}_q^2 \setminus \bigcup_{i \in \mathcal{I}} \bigcup_{k=1}^K \bigcup_{\ell \in \mathcal{B}_{i,k}} \ell.$$

1. *Every cover of X_{pool} by at most q affine lines contains at least $(1 - \rho)q$ lines through a common projective point.*
2. *For every affine line ℓ , if only exact occurrences of ℓ are ignored in the pool deletions, at least $\delta q/2$ points of ℓ remain.*
3. *For every finite point p and affine line ℓ not through p , at most five groups i have a candidate $P_{i,k}$ containing a point $y \in \ell$ whose join to p has direction in S_i .*

Proof. We verify the hypotheses of Lemma 3.2 for the full pool, which gives the first two assertions. The third follows from a separate union bound over finite points and lines.

Treat each candidate shift as one independent deletion trial, listing the lines of $\mathcal{B}_{i,k}$. There are $KN_q = O_{s,K}(q)$ trials, each listing $m = (s - 1)^2$ lines. We may take the shape family to be all $q(q + 1)$ ambient affine lines. Each shape has size q , and two distinct line shapes intersect in at most one point. Thus the shape and trial bounds in Lemma 3.2 hold with a sufficiently large fixed integer D depending only on s, K .

A candidate's deleting set is a uniform translate of a fixed union of m lines. Its deletion probability at a fixed point is independent of the point and is at most m/q . Independence of the shifts makes the overall survival probability δ_q uniform on $\mathbb{F}_q^2$. For sufficiently large q , $m/q \leq 1/2$ and $N_q \leq 2q/(s - 1)$. Using $\log(1 - t) \geq -2t$ for $0 \leq t \leq 1/2$, we obtain

$$\delta_q \geq (1 - m/q)^{KN_q} \geq \exp(-2KN_q m/q) \geq \exp(-4K(s - 1)).$$

Fix $\delta = \exp(-4K(s - 1))$ and $\rho = \delta/10$.

For completeness, the exact-line hypothesis also has a uniform bound. A fixed line ℓ of direction d can occur only in candidates of groups with $d \in S_i$, of which there are at most two. In each such candidate, there are $s - 1$ translated lines in this direction. Each fixed line has probability $1/q$ of becoming ℓ under the uniform shift. Therefore the sum, over all trials, of the probabilities of listing ℓ is at most

$$\frac{2K(s - 1)}{q}. \tag{74}$$

Increasing the fixed D if necessary verifies every hypothesis of Lemma 3.2. Its conclusions give the first two properties.

To prove the third, fix $p \notin \ell$. For one candidate in group i , there are at most $s - 1$ possible points $y \in \ell$ whose join to p has a direction in S_i : each such direction contributes at most one intersection with ℓ . A fixed point belongs to a uniform translate of W_i with probability s^2/q^2 . The probability for this candidate is thus at most

$$\frac{(s - 1)s^2}{q^2}.$$

If six distinct groups meet the condition, choose one witnessing candidate from each. There are at most $(KN_q)^6 = O_{s,K}(q^6)$ choices, and their shifts are independent. For the fixed pair (p, ℓ) , the probability of six such groups is therefore $O_{s,K}(q^{-6})$. There are at most $q^2 q(q + 1) = O(q^4)$ pairs (p, ℓ) . The union bound gives a total failure probability $O_{s,K}(q^{-2})$, proving the claim. $\square$

6.3 Selection and transfer of the deletion estimates

The estimates just proved concern independent candidates. We can now select compatible copies, using only inclusions to transfer the estimates to the selected deletion set. By Theorem 4.1, with probability tending to one the same pool admits one candidate per group with no conflicts. Intersect this event with those of Lemma 6.2. No independence between these events is needed. For every sufficiently large odd integer n , their intersection has positive probability, so fix a pool in the intersection and a conflict-free selection. Denote the selected sets and shifts by

$$P_i = b_i + M_i(\mathbb{F}_s^2), \quad Y_i = b_i + M_i Y.$$

Conflict avoidance gives the useful implication

$$i \neq j, \quad \text{dir}(\ell) \in D_i, \quad \ell \cap P_i \neq \emptyset \implies \ell \cap P_j = \emptyset. \quad (75)$$

In particular the P_i are pairwise disjoint: a common point would give a zero difference, which belongs to every conflict direction, and each D_i is nonempty.

Let B be the union, as a set of lines, of the deletion lines of the selected candidates, and put

$$X = \mathbb{F}_q^2 \setminus \bigcup_{\ell \in B} \ell. \quad (76)$$

Then $X \supseteq X_{\text{pool}}$. Moreover, if $\ell \notin B$, every selected deletion line belongs to the pool list after all exact occurrences of ℓ have been omitted. Hence X contains the survivor set of that modified pool process. This inclusion is deterministic: it holds even if ℓ occurs in an unselected candidate and even though the selected shifts depend on the whole pool. In particular,

$$\ell \notin B \implies |X \cap \ell| \geq \frac{\delta q}{2}. \quad (77)$$

The third property of Lemma 6.2 also remains available for the selection, since it was established for all candidates in the pool.

This completes the placement: the copies are separated, covers of the ordinary survivors by at most q lines contain an almost pencil, every undeleted line contains at least $\delta q/2$ survivors, and the bound of five groups remains valid. The order of choices is s, Y , then the selection constant and K , then the deletion constants D, δ, ρ , and finally $q = s^n$. Only after fixing these constants do we require $q = s^n$ to be sufficiently large. All further numerical inequalities below involve these fixed constants and can be imposed by enlarging this final threshold. Thus the construction is available for every sufficiently large odd integer n , with the same s, Y ; its pools and selected shifts may depend on n . The constant used for the ambient deletion lemma may depend on K , but it changes no hypothesis of the already applied selection theorem. A small value of δ only increases the final lower threshold on q .

7 The hypergraph and its covering number

The placement supplies the ordinary points X and the exceptional copies Y_i . We now turn them into edges. Ordinary edges retain all their affine-line labels. In an exceptional copy, the directions S_i instead use line labels read from $f(Y)$; all remaining directions retain their ordinary labels. We will first check that every pair of edges still meets, then use the rigidity of $f(Y)$ to rule out covers of size q .

7.1 Vertices, edges, and intersection

Set $r = q + 1$. There is one vertex part V_d for each $d \in \mathbb{P}^1(\mathbb{F}_q)$. For every ambient affine line ℓ of direction d , place an ordinary vertex o_ℓ in V_d . If $d \in S_i$, let $t_i(d) \in \mathbb{F}_s^*$ be the unique subfield direction whose image under M_i is d . For each of the s subfield affine lines L of direction $t_i(d)$, place a special vertex $v_{i,d,L}$ in V_d . These subfield lines are read in the new coordinates used for $f(Y)$. All vertices are formal labels, tagged to be distinct between parts, between ordinary and special types, and between different groups. In particular each V_d is finite, and the parts are pairwise disjoint.

Write $\ell_d(z)$ for the ambient line of direction d through z , and $L_t(w)$ for the subfield line of direction t through $w \in \mathbb{F}_s^2$. Each $x \in X$ defines the ordinary edge

$$e_x = \{o_{\ell_d(x)} : d \in \mathbb{P}^1(\mathbb{F}_q)\}.$$

For each $i \in \mathcal{I}$ and $w \in Y$, put $y_{i,w} = b_i + M_i w$ and define an exceptional edge by selecting, in part V_d , the vertex

$$\begin{cases} v_{i,d,L_{t_i(d)}(f(w))}, & d \in S_i, \\ o_{\ell_d(y_{i,w})}, & d \notin S_i. \end{cases} \quad (78)$$

Denote this edge by $e_{i,w}$, and let

$$H = \{e_x : x \in X\} \cup \{e_{i,w} : i \in \mathcal{I}, w \in Y\}.$$

Thus an exceptional edge changes labels in the assigned directions, without introducing any new part. Every edge contains exactly one vertex from every part, so H is $(q + 1)$ -partite and $(q + 1)$ -uniform. It is nonempty, because $Y \neq \emptyset$.

Figure 1 shows how to read an exceptional edge from its parameter w . The deletion lines were defined from its ordinary base point $y_{i,w}$, while its special labels use $f(w)$. The two axis parallel classes are preserved by f , which joins these descriptions in the intersection proof.

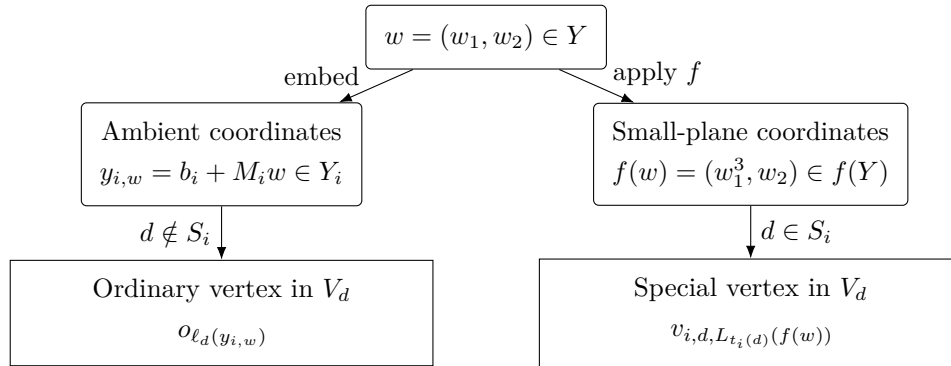


Figure 1: The two label rules for the vertex of $e_{i,w}$ in part V_d . The left branch uses the embedded point $b_i + M_i w$; the right branch uses $f(w) = (w_1^3, w_2)$ in the small plane. For each $d \in A_i$, two parameters give the same ordinary label exactly when their images under f lie on the same corresponding horizontal or vertical line. The diagram records maps and label choices, rather than geometric positions in the finite plane.

Proposition 7.1. *The hypergraph H is intersecting. Furthermore, all the edges in its displayed parametrization are distinct.*

Proof. There are four types of pairs, according to whether their edges are ordinary or exceptional and, in the latter case, whether their group indices agree.

Two ordinary edges with distinct base points share the ordinary label of their join. Next, an exceptional base point $y \in Y_i$ lies on a deletion line in every direction of the nonempty set S_i , so $y \notin X$. Its join with $x \in X$ cannot have direction in S_i : otherwise that join would be a selected deletion line containing x . Its ordinary label therefore belongs to both the exceptional edge and e_x .

For exceptional edges from distinct groups i, j , the base points are distinct by (75). That same separation condition ensures that their join has direction outside both D_i and D_j , and hence outside $S_i \cup S_j$. The two edges share its ordinary label.

Finally, consider $e_{i,w}$ and $e_{i,w'}$ with $w \neq w'$. The transformed points $f(w), f(w')$ are distinct. If their join has finite nonzero slope t , the exceptional edges share the special label of this subfield line in the part of direction $M_i(t)$. If their join is horizontal or vertical, the relevant original coordinates are equal as well, because f preserves these two parallel classes bijectively. The original base points then share an ordinary axis-direction label, in a direction of A_i . This completes the intersection proof.

We also check that the parameters count distinct edges, as required when we count exceptional edges below. Two ordinary labels in nonparallel directions determine the ordinary base point, so ordinary edges are distinct. An exceptional edge has special labels and cannot equal an ordinary edge. Within one group, the special labels in any two distinct directions of S_i determine $f(w)$, since the corresponding subfield lines are nonparallel. As $|S_i| = s - 1 \geq 2$, equal exceptional edges within a group would imply $w = w'$. Finally, any special label tagged by group i is absent from every edge of a different group. Exceptional edges in distinct groups are therefore distinct too. $\square$

7.2 Covering vertices in the small-plane coordinates

We identify the positions of the exceptional edges of group i with $f(Y)$, via $e_{i,w} \leftrightarrow f(w)$. The action of a covering vertex means the subset of these positions whose edges contain that vertex. The next lemma translates every such action into one of the two resources in Lemma 5.1: a subfield line or an individual point.

Lemma 7.2. *Fix a group i .*

1. *A special vertex of group i covers the positions on one subfield line. A special vertex of another group covers none.*
2. *An ordinary vertex of direction in S_i covers none. An ordinary vertex of direction in A_i , if it covers any positions, covers them on a single horizontal or vertical line in the f coordinates.*
3. *An ordinary vertex of direction outside $S_i \cup A_i$ covers at most one position.*

In particular, the positions covered by any one vertex are contained in a subfield affine line.

Proof. The assertions for special vertices and altered ordinary directions follow directly from (78). For an axis direction, undoing b_i, M_i fixes one original coordinate, so applying f gives the corresponding horizontal or vertical line. A line in a direction outside $S_i \cup A_i$ cannot contain two points of P_i , since the join directions of that embedded subplane are exactly $S_i \cup A_i$. It therefore covers at most one exceptional position. A singleton can be included in an arbitrary subfield line through it, and an empty set requires no covering resource. $\square$

7.3 The final cover argument

Completion of the proof of Theorem 1.1. We prove that H has no vertex cover of size at most q . Suppose to the contrary that C is such a cover. Its ordinary vertices, regarded as affine lines, must cover X : special vertices belong to no ordinary edge. These lines also cover X_{pool} . Lemma 6.2 therefore gives a projective point p such that at least $(1 - \rho)q$ ordinary vertices of C are lines through p . The line-survival bound will force every undeleted member of this pencil into C . We then count the vertices left for exceptional edges, separating centers at infinity, centers inside one of the sets P_i , and finite centers outside them all.

Every undeleted line in this pencil is forced. We claim that

$$p \in \ell, \quad \ell \notin B \quad \implies \quad o_\ell \in C, \quad (79)$$

where incidence with an infinite p means that ℓ has the corresponding direction. If o_ℓ were missing, all other lines of C through p would cover at most one affine point of ℓ . There are at most ρq vertices outside the large pencil subcollection. Each of their ordinary lines covers at most one point of ℓ , and their special vertices cover no ordinary edge. Consequently

$$|X \cap \ell| \leq 1 + \rho q.$$

This contradicts (77), since $\delta q/2 > 1 + \rho q$ for sufficiently large q and $\rho = \delta/10$. At infinity the pencil lines are parallel, so the possible contribution of one central point is absent; the same upper bound still applies.

Let F_p be the set of ordinary vertices forced by (79), and call the vertices of $C \setminus F_p$ the additional vertices. The deleted pencil lines determine how many additional vertices the assumed budget $|C| \leq q$ permits.

An infinite center. Suppose p is the point at infinity in direction d . Put

$$I_d = \{i \in \mathcal{I} : d \in S_i\}, \quad z = |I_d| \in \{1, 2\}.$$

Only these groups contribute deletion lines in direction d , and each contributes $s - 1$, by Lemma 6.1. At most $z(s - 1)$ of the q lines in this parallel class belong to B . Thus

$$|C \setminus F_p| \leq z(s - 1).$$

None of the forced vertices covers any exceptional edge of a group in I_d , because those groups use special vertices in direction d .

If $z = 1$, all positions of this group's $f(Y)$ must therefore be covered by at most $s - 1$ additional vertices. By Lemma 7.2, their effects can be included in at most $s - 1$ subfield lines. This contradicts Lemma 5.1.

If $z = 2$, the two groups are $1, *$, by (72). An additional vertex can now meet both groups, so we use the weighted line-and-point budget instead of counting one line per vertex. For each of these groups let a_i count the additional vertices that cover at least one of its positions and are either special vertices of this group or ordinary vertices in its axis directions A_i . Let b'_i count the other additional vertices covering a position of the group. By Lemma 7.2, the latter vertices each cover only a singleton. The former supply subfield lines. The weighted conclusion of Lemma 5.1 gives

$$a_i + \frac{b'_i}{2} \geq s \quad (i = 1, *). \quad (80)$$

Each additional vertex contributes at most one to the sum of the two left sides. A special vertex is confined to its tagged group. An ordinary axis-direction vertex counted in a_i is a line meeting P_i with direction in $A_i \subseteq D_i$; by (75) it cannot meet the other group's P_j . It therefore contributes nothing to that other group, even as a singleton. Any vertex of neither exclusive type contributes at most $1/2$ to each group. This accounts for every type of additional vertex, including those covering neither group. Summing (80), we obtain

$$2s \leq a_1 + \frac{b'_1}{2} + a_* + \frac{b'_*}{2} \leq |C \setminus F_p| \leq 2(s-1),$$

a contradiction.

A finite center inside an embedded configuration. Now let p be finite, and write

$$h_p = |\{\ell \in B : p \in \ell\}|.$$

There are $q+1$ affine lines through p , so

$$|F_p| = q+1 - h_p, \quad |C \setminus F_p| \leq h_p - 1. \quad (81)$$

The loss of one in the second inequality comes from the $q+1$ members of a finite pencil and the assumed cover size q . If $h_p = 0$, the forced vertices already exceed the permitted size of C . We may therefore assume $h_p \geq 1$ in both finite-center cases.

Suppose $p \in P_i$, for the necessarily unique such group. Every deletion line through p comes from this group: a line from another group would contradict (75), since its direction belongs to that other group's conflict set. There is at most one such line in each direction of S_i , and hence

$$h_p \leq s-1. \quad (82)$$

Let $w_p = M_i^{-1}(p - b_i) \in \mathbb{F}_s^2$, whether or not $w_p \in Y$. We account for the exceptional positions of this group covered by F_p : they will all lie on two nonparallel lines in the new coordinates. For a position with base point $y \neq p$, its join with p has a direction in $S_i \cup A_i$. A direction in S_i cannot cover this edge by an ordinary vertex. Any actual forced coverage must therefore lie along one of the two axis-direction lines through p . A position based at p itself, if present, lies on both these axes as well.

Since f preserves the two coordinate parallel classes, all these covered positions are contained, in the new coordinates, in the horizontal and vertical lines through $f(w_p)$. The remaining positions are covered by the additional vertices; each can be included in one subfield line by Lemma 7.2. Equations (81) and (82) would thus give a cover of $f(Y)$ by at most

$$2 + (h_p - 1) \leq s$$

lines containing the two nonparallel coordinate lines. These are legitimate covering resources whether or not their ordinary labels belong to F_p : their union contains every position covered by F_p . Removing any duplicate replacement lines preserves these two distinct lines and can only decrease the count. This contradicts Lemma 5.1.

A finite center outside all embedded configurations. It remains to consider $p \notin \bigcup_{i \in \mathcal{I}} P_i$. Here we use the five points on each deletion line, rather than the small-plane cover obstruction. We will find $5h_p$ exceptional edges missed by F_p , while each additional vertex covers at most five of them.

Each group contributes at most one deletion line through p . Indeed, two distinct such lines cannot be parallel. After undoing b_i, M_i , their subfield affine equations have distinct subfield directions. Solving these two equations uses a nonsingular linear system over $\mathbb{F}_s$, so their intersection lies in $\mathbb{F}_s^2$. This would place p in P_i , a contradiction. Also, deletion lines from different groups cannot coincide, by (75).

The h_p deletion lines through p therefore come from h_p distinct groups. From each such group i , choose five distinct points of Y_i on its contributing line, possible by Lemma 6.1. These give $5h_p$ distinct exceptional edges, by Proposition 7.1. None is covered by a forced ordinary vertex: its unique join line to p is precisely its contributing deletion line, whose direction is altered for that group.

Any one additional vertex covers at most five of these selected edges. For a special vertex this follows because it belongs to only one group. An ordinary line through p covers none, by the same unique-join and altered-direction argument. Finally, let an ordinary vertex o_ℓ , with ℓ not through p , cover some selected edges. The line ℓ meets each contributing line in at most one point, so o_ℓ covers at most one selected edge from each contributing group. Every such group has a selected point $y \in Y_i \cap \ell$ whose join to p has direction in S_i . The third part of Lemma 6.2, which applies to all candidates and hence to the selection, bounds the number of these groups by five. This proves the asserted bound for every additional vertex.

By (81), at most $h_p - 1$ additional vertices are available. They can cover at most $5(h_p - 1)$ of the $5h_p$ selected edges, a final contradiction.

We have ruled out every possible projective center p , and hence every vertex cover of size at most q . Therefore $\tau(H) \geq q + 1$. Conversely, any edge of the nonempty intersecting hypergraph H is itself a vertex cover and has size $q + 1$. It follows that

$$\tau(H) = q + 1.$$

The parameter choices in Section 6 give this hypergraph for every sufficiently large odd integer n , with $q = s^n$, after fixing any sufficiently large prime $s \equiv 2 \pmod{3}$ and a successful Y . Choose one such Y for each eligible s , and let $n_0(s) \geq 3$ exceed all the resulting degree thresholds. This proves Theorem 1.1. Since there are arbitrarily large odd integers, it also gives infinitely many ranks $r = q + 1$. $\square$

References

- [1] Aida Abiad, Frederik Garbe, Xavier Povill, and Christoph Spiegel. Infinitely many counterexamples to a conjecture of Lovász. [arXiv:2506.21286v2](#), 2025. Version 2, 14 July 2025.
- [2] Ahmad Abu-Khazneh, János Barát, Alexey Pokrovskiy, and Tibor Szabó. A family of extremal hypergraphs for Ryser’s conjecture. *Journal of Combinatorial Theory, Series A*, 161:164–177, 2019. [doi:10.1016/j.jcta.2018.07.011](#).
- [3] Ron Aharoni. Ryser’s conjecture for tripartite 3-graphs. *Combinatorica*, 21(1):1–4, 2001. [doi:10.1007/s004930170001](#).
- [4] Ron Aharoni, János Barát, and Ian M. Wanless. Multipartite hypergraphs achieving equality in Ryser’s conjecture. *Graphs and Combinatorics*, 32(1):1–15, 2016. [arXiv:1409.4833](#).
- [5] Antal Balog and Endre Szemerédi. A statistical theorem of set addition. *Combinatorica*, 14(3):263–268, 1994. [doi:10.1007/BF01212974](#).
- [6] Darcy Best and Ian M. Wanless. What did Ryser conjecture?, 2018. [arXiv:1801.02893](#).
- [7] Anurag Bishnoi, Shagnik Das, Patrick Morris, and Tibor Szabó. Ryser’s conjecture for t -intersecting hypergraphs. *Journal of Combinatorial Theory, Series A*, 179:105366, 2021. [doi:10.1016/j.jcta.2020.105366](#).
- [8] A. Blokhuis, A. E. Brouwer, and T. Szőnyi. Covering all points except one. *Journal of Algebraic Combinatorics*, 32:59–66, 2010. [doi:10.1007/s10801-009-0204-1](#).
- [9] Jean Bourgain, Nets Katz, and Terence Tao. A sum-product estimate in finite fields, and applications. *Geometric and Functional Analysis*, 14(1):27–57, 2004. [doi:10.1007/s00039-004-0451-1](#).
- [10] A. E. Brouwer and A. Schrijver. The blocking number of an affine space. *Journal of Combinatorial Theory, Series A*, 24(2):251–253, 1978. [doi:10.1016/0097-3165\(78\)90013-4](#).
- [11] Alexander Clow, Penny Haxell, and Bojan Mohar. A counterexample to a conjecture of Lovász. *Combinatorica*, 46:26, 2026. Article 26. [doi:10.1007/s00493-026-00220-3](#).
- [12] Louis DeBiasio, Yigal Kamel, Grace McCourt, and Hannah Sheats. Generalizations and strengthenings of Ryser’s conjecture. [arXiv:2009.07239v3](#), 2021. Version 3, 3 November 2021.
- [13] P. Erdős, A. Gyárfás, and L. Pyber. Vertex coverings by monochromatic cycles and trees. *Journal of Combinatorial Theory, Series B*, 51(1):90–95, 1991. [doi:10.1016/0095-8956\(91\)90007-7](#).
- [14] Jacob Fox and Benny Sudakov. Dependent random choice. *Random Structures & Algorithms*, 38(1–2):68–99, 2011. [arXiv:0909.3271](#).
- [15] Nevena Francetić, Sarada Herke, Brendan D. McKay, and Ian M. Wanless. On Ryser’s conjecture for linear intersecting multipartite hypergraphs. *European Journal of Combinatorics*, 61:91–105, 2017. [doi:10.1016/j.ejc.2016.10.004](#).
- [16] David A. Freedman. On tail probabilities for martingales. *The Annals of Probability*, 3(1):100–118, 1975. [doi:10.1214/aop/1176996452](#).

- [17] W. T. Gowers. A new proof of Szemerédi’s theorem for arithmetic progressions of length four. *Geometric and Functional Analysis*, 8(3):529–551, 1998. [doi:10.1007/s000390050065](#).
- [18] P. E. Haxell. A condition for matchability in hypergraphs. *Graphs and Combinatorics*, 11(3):245–248, 1995. [doi:10.1007/BF01793010](#).
- [19] P. E. Haxell. A note on vertex list colouring. *Combinatorics, Probability and Computing*, 10(4):345–347, 2001. [doi:10.1017/S0963548301004758](#).
- [20] P. E. Haxell and A. D. Scott. On Ryser’s conjecture. *The Electronic Journal of Combinatorics*, 19(1):P23, 2012. [doi:10.37236/1175](#).
- [21] P. E. Haxell and A. D. Scott. A note on intersecting hypergraphs with large cover number. *The Electronic Journal of Combinatorics*, 24(3):P3.26, 2017. [doi:10.37236/6460](#).
- [22] John Robert Henderson. *Permutation Decompositions of $(0,1)$ -Matrices and Decomposition Transversals*. PhD thesis, California Institute of Technology, 1971. [doi:10.7907/J1Z1-SK19](#).
- [23] Robert E. Jamison. Covering finite fields with cosets of subspaces. *Journal of Combinatorial Theory, Series A*, 22(3):253–266, 1977. [doi:10.1016/0097-3165\(77\)90001-2](#).
- [24] Zoltán Király and Lilla Tóthmérész. On Ryser’s conjecture for t -intersecting and degree-bounded hypergraphs. *The Electronic Journal of Combinatorics*, 24(4):P4.40, 2017. [doi:10.37236/6448](#).
- [25] Toufik Mansour, Chunwei Song, and Raphael Yuster. A comment on Ryser’s conjecture for intersecting hypergraphs. *Graphs and Combinatorics*, 25(1):101–109, 2009. [arXiv:0709.3138](#).
- [26] Luka Milićević. Covering complete graphs by monochromatically bounded sets. [arXiv:1705.09370v1](#), 2017. Version 1, 25 May 2017.
- [27] OpenAI. Balanced counterexamples to Ryser’s conjecture at prime orders. OpenAI Math Release preprint [OAI:Balanced-Counterexamples-to-Rysers-Conjecture-at-Prime-Orders-September-27-2026](#), 2026.
- [28] Giorgis Petridis. New proofs of Plünnecke-type estimates for product sets in groups. *Combinatorica*, 32(6):721–733, 2012. [doi:10.1007/s00493-012-2818-5](#).
- [29] Imre Z. Ruzsa. An analog of Freiman’s theorem in groups. In *Structure theory of set addition*, volume 258 of *Astérisque*, pages 323–326. Société Mathématique de France, 1999. [Primary source](#).
- [30] Zsolt Tuza. Ryser’s conjecture on transversals of r -partite hypergraphs. *Ars Combinatoria*, 16B:201–209, 1983. [Publisher record](#).
- [31] Patrick White. Tuza’s Ryser-conjecture claim for four-partite hypergraphs with matching number two. [arXiv:2609.14281v1](#), 2026. Version 1, 13 September 2026.